SMART BUILDINGS

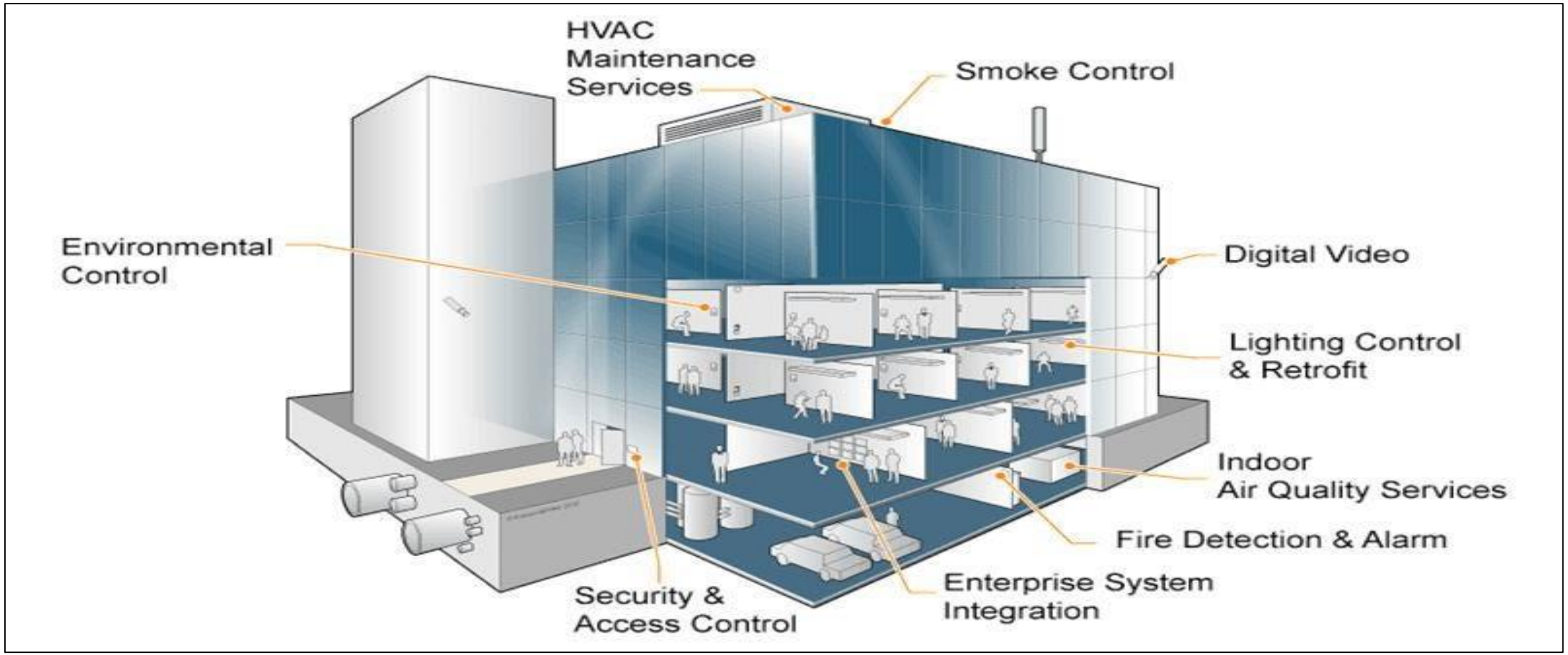
Evolution of Buildings Throughout History

SMART BUILDINGS - GENERAL

- Smart buildings are structures equipped with various sensors, systems, and technologies to optimize efficiency, comfort, security, and sustainability.
- These buildings leverage data and automation to enhance operations and improve the overall experience for occupants.
- A smart building involves the installation and use of advanced and integrated building technology systems.
- They provide actionable information about a building or space within a building to allow the building owner or occupant to manage the building.

SMART BUILDINGS - GENERAL

- The driving forces for smart buildings are economics, energy, and technology.
- Smart buildings leverage mainstream information technology infrastructure and take advantage of existing and emerging technology such as Digital Controls and Networking, Internet of Things (IoT), Data Analytics and AI.
- For developers and owners, smart buildings increase the value of a property.



Typical arrangements in a smart building

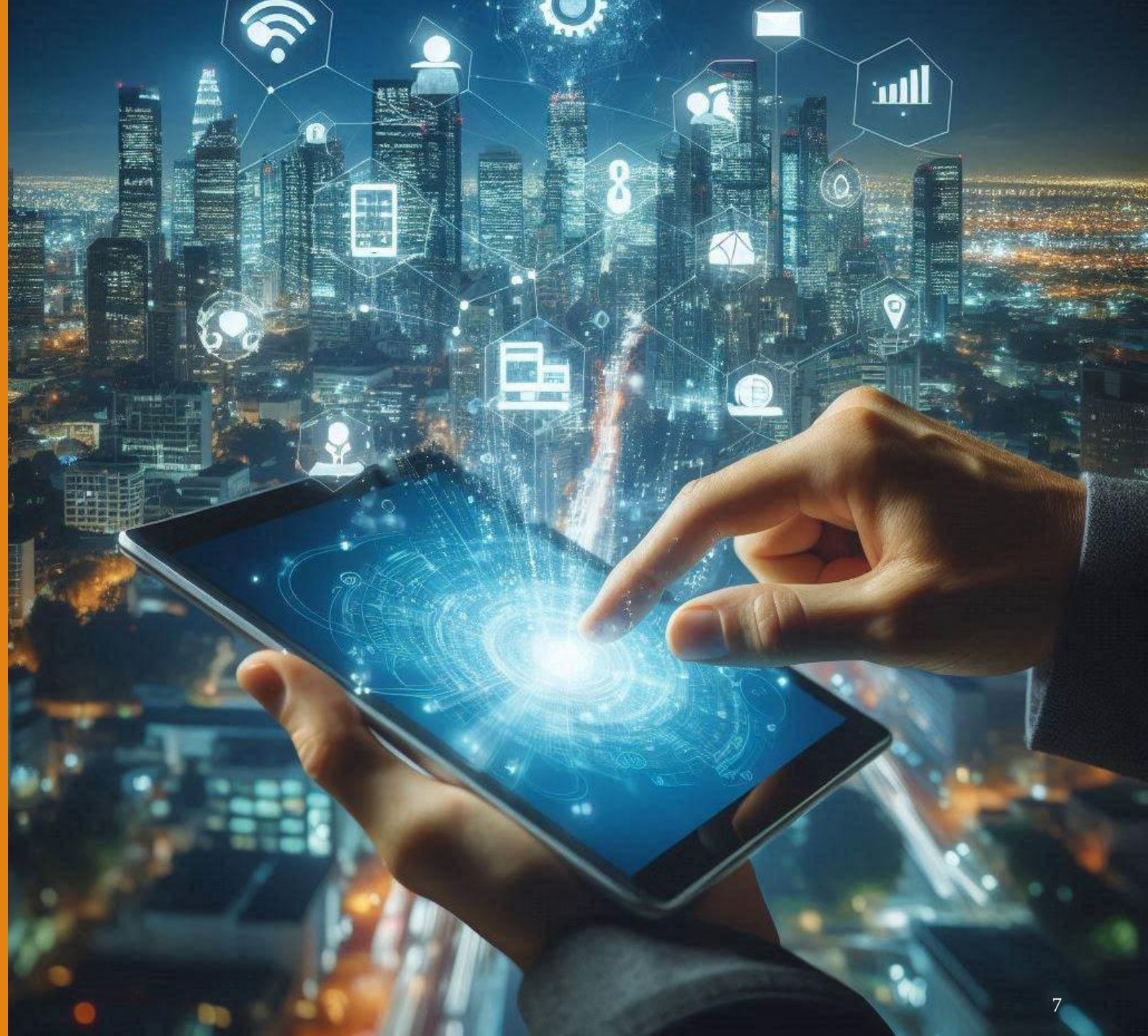
Source: https://medium.com/s-o-t-a/take-a-look-around-youre-in-a-smart-building-4023e0b9f1ba

SUMMARY

- Discussion on Evolution of buildings, from prehistoric times to current trends
- Definition of smart building and discussion
- Typical arrangement in smart buildings

https://www.youtube.com/watch?v=8NzsQw46kDI

LECTURE – 08
HVAC SYSTEM



IMPORTANCE OF HVAC SYSTEM

- HVAC stands for Heating, Ventilation, and Air Conditioning. It's a system designed to control the environment within a building or a vehicle to ensure thermal comfort and air quality.

- A typical system brings in outside air, mixes it with air returned from or exiting the system, filters the air, passes it through a heating or cooling coil to a required temperature, and distributes the air to the various sections of a building.

IMPORTANCE OF HVAC SYSTEM

- The HVAC system not only makes the building comfortable, healthful, and livable for its occupants, it manages a substantial portion of energy usage and related costs for the building.
- The HVAC system is also critical in controlling smoke in the event of a fire.
- An HVAC system having a single control thermostat serves one zone of a “thermal load.” Most large buildings have multiple zone systems, with air supplied to each zone specifically addressing its needs and thermal load.

COMPONENTS OF HVAC

- HVAC systems can be very complex, consisting of many components.

- The major components include boilers, chillers, air-handling units (AHUs), air terminal units (ATUs), and variable air volume equipment (VAV).

COMPONENTS OF HVAC

Boilers

Boilers are used to heat air.

Boilers heat air in the following manner: a fuel (typically propane or natural gas) is combusted, and the resulting heat is used to heat water. The hot water or steam is piped through the building to radiator units where air is forced over them, moving heated air through the ducts and into the rooms.

Commercial boilers

Source: James Sinopli, Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

Boilers are available in two main categories: conventional units and condensing units.

Type of Boilers

Conventional Units are typically made of materials that cannot handle the corrosive properties of the condensing gases, and therefore that heat becomes waste.

Condensing Units allow the water vapor produced during the combustion of whatever hydrocarbon fuel is used to produce heat to condense. Condensing units typically have efficiencies of over 90% and are more energy efficient than conventional units.

Components of HVAC

Chillers

Chillers, or air conditioners, utilize heat exchanges and circulate fluid or gas to cool the air that is passed through the unit

Cooling Tower

Source: James Sinopli, Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

COMPONENTS OF HVAC

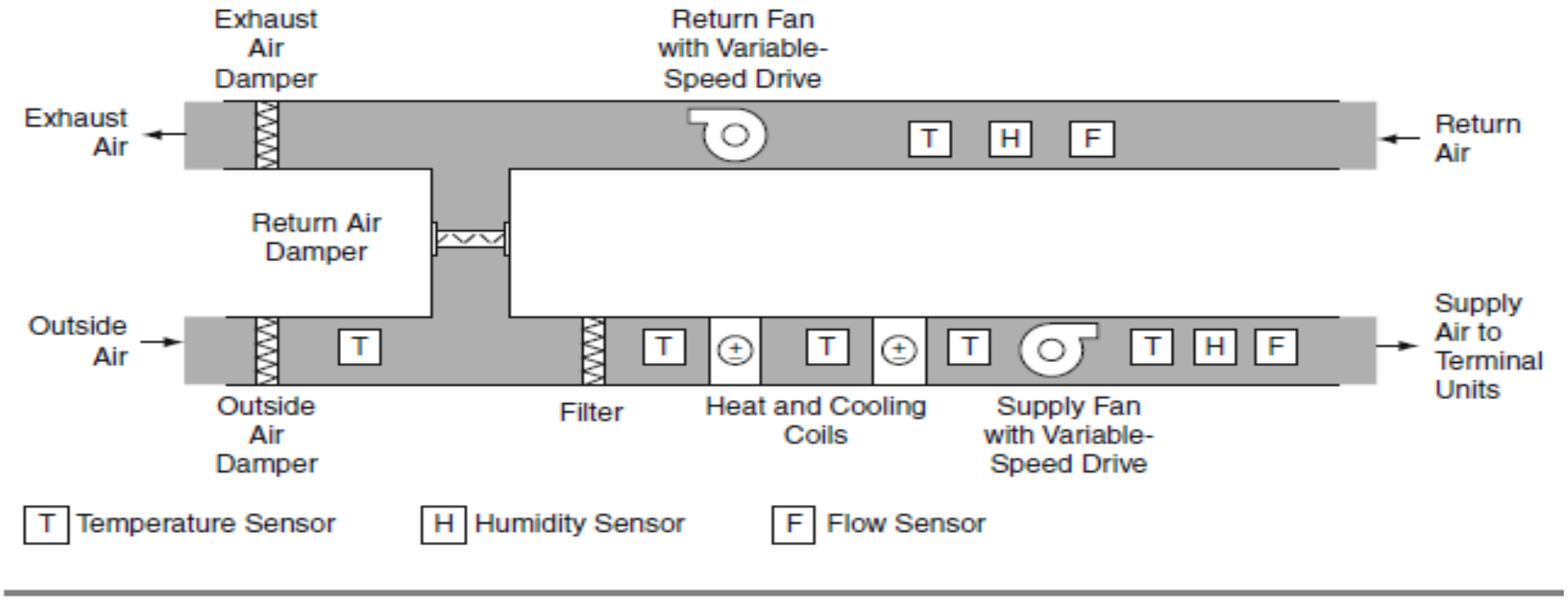
Air Handling Unit

- Air-handling units (AHUs) provide warm or cool air to different parts of a building, using chilled water to cool the air or steam or hot water to heat the air.

- An air handler is usually a metal box containing a blower, heating and/or cooling elements, filter racks or chambers, sound attenuators, and dampers.

COMPONENTS OF HVAC

Air Handling Unit



Source: James Sinopli, Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

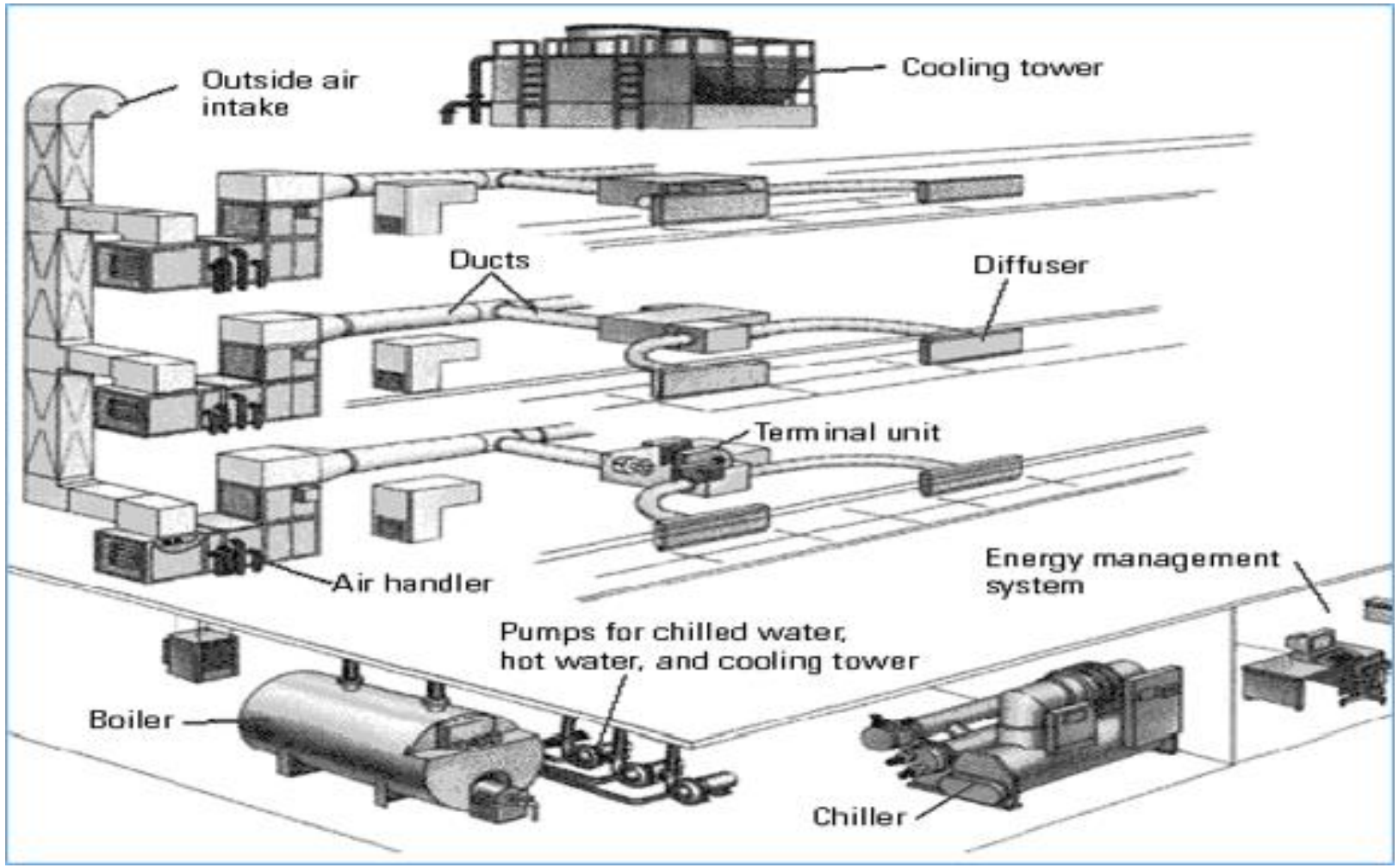
COMPONENTS OF HVAC

Air Terminal Unit

- Air terminal units (ATUs) address specific HVAC thermal loads or zones. Thermal loads in a space can consist of exterior loads (outside air temperatures increasing or decreasing) and interior loads (people, lighting, computers, and other sources).

- ATUs compensate for these thermal loads and zones by varying the air temperature, varying the air volume, or doing both. While constant air volume (CAV) systems provide air at a variable temperature and constant flow rate, variable air volume (VAV) systems provide air at a constant temperature and regulate the room temperature by changing the flow rate of the air into the room.

COMPONENTS OF HVAC SYSTEM



Source: James Sinopli, Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

HVAC EFFICIENCY

Air conditioner efficiency: Is essentially a ratio of the cooling capacity of the unit versus its required power input and is usually measured using EER or SEER rating:

Energy efficiency ratio (EER): is a measurement of the energy efficiency of an air conditioner at a specific outdoor temperature (95 degrees F). The system is more efficient when its EER rating is higher.

$$EER\,Rating = \frac{Cooling\,Output\,(in\,BTU/hr)}{Power\,Input\,(Max.Wattage)}$$

EER RATING

Example: We are testing portable air conditioners at the test conditions (95°F, 80°F, 50% humidity). The maximum wattage the unit runs on is 1,000W. We now measure that the total cooling output when the AC is running at 1,000W is 12,000 BTU/hr. Basically, we have a 12,000 BTU air conditioner. Here's how the EER rating is calculated:

EER Rating = 12,000 BTU/hr ÷ 1,000 W = **12**

- At different temperatures and relative humidity, the air conditioner will have different performances.
- In short, the EER rating test conditions are the perfect conditions.
- If you were to measure the energy efficiency of an air conditioner at let's say 110°F outdoor temperature, 78°F indoor temperature, and 60% relative humidity the energy efficiency will fall.

Source: https://learnmetrics.com/eer-vs-seer/

SEER RATING

Seasonal energy efficiency ratio (SEER): provides a more accurate measure of a unit's efficiency because it is taken as an average of various operating conditions and seasonal temperatures. It is designed to give the most accurate representation of actual operating conditions for the air conditioner. Because it is based on seasonal external temperatures SEER ratings depend on geographical location and thereby differ by area.

$$SEER\,Rating = \frac{1 \times EER_{100\%} + 42 \times EER_{75\%} + 45 \times EER_{50\%} + 12 \times EER_{25\%}}{100}$$

SEER RATING

Example: Let's say that we have a 12,000 BTU air conditioner with a 12 EER rating from the previous example. Here are theoretical measurements of energy efficiency at 25%, 50%, 75%, and 100% wattage:

- 4,000 BTU at 25% load (250W). That yields a 16 EER25% rating.
- 7,500 BTU at 50% load (500W). That yields a 15 EER50% rating.
- 10,500 BTU at 75% load (750W). That yields a 14 EER75% rating.
- 12,000 BTU at 100% load (1,000W). That yields a 12 EER100% rating.

With these partial and full EER loads, we can calculate the SEER rating

SEER Rating = $(1\times12 + 42\times14 + 45\times15 + 12\times16) / 100 = 14.67$

In general, the SEER rating is always higher than the full load EER rating.

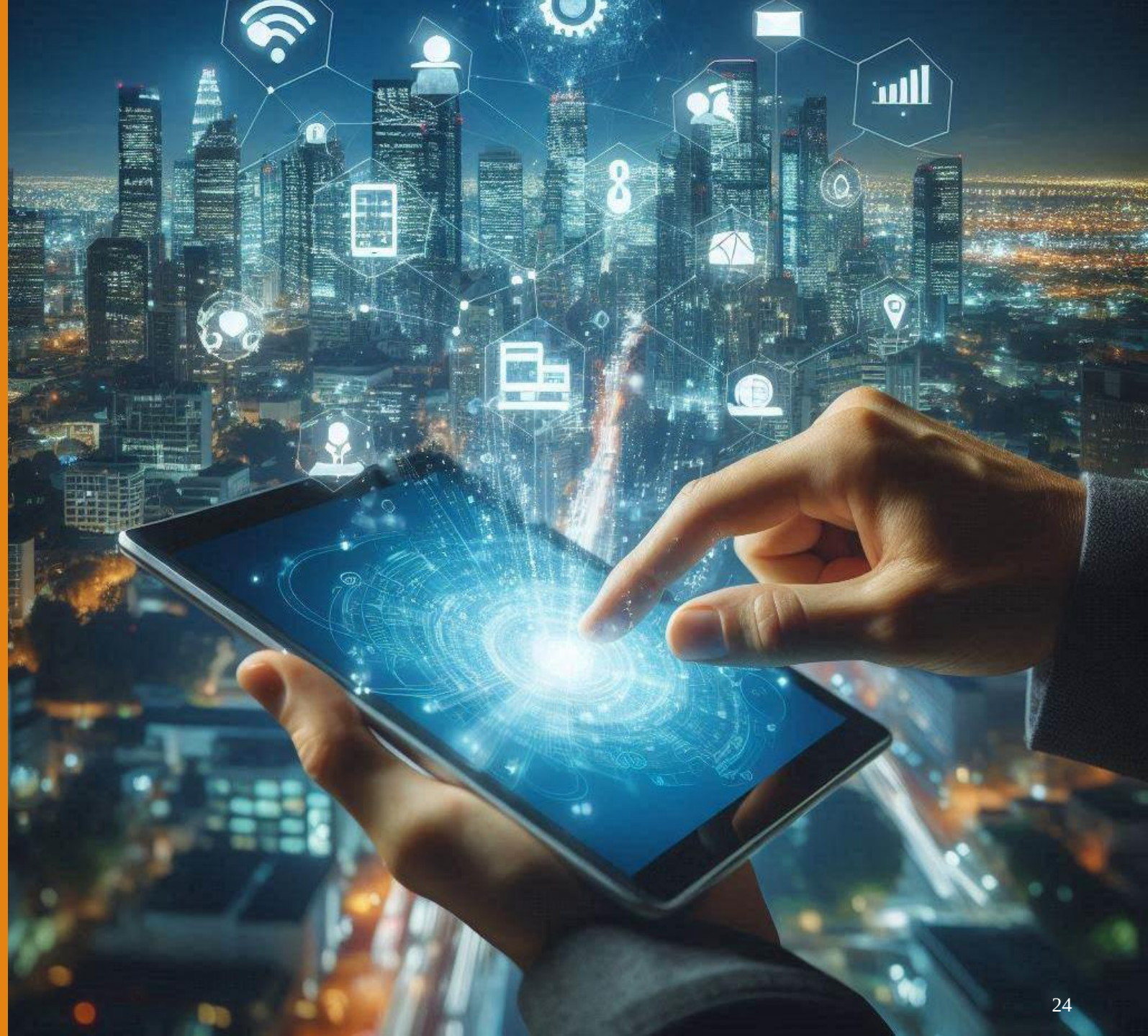
Source: https://learnmetrics.com/eer-vs-seer/

SUMMARY

- HVAC system and its importance
- Various components of HVAC.
- Typical HVAC system
- HVAC efficiency ratings

https://www.youtube.com/watch?v=tRB9gyXitGk

LECTURE – 09
LIGHTING CONTROL SYSTEM



LIGHTING CONTROL SYSTEM

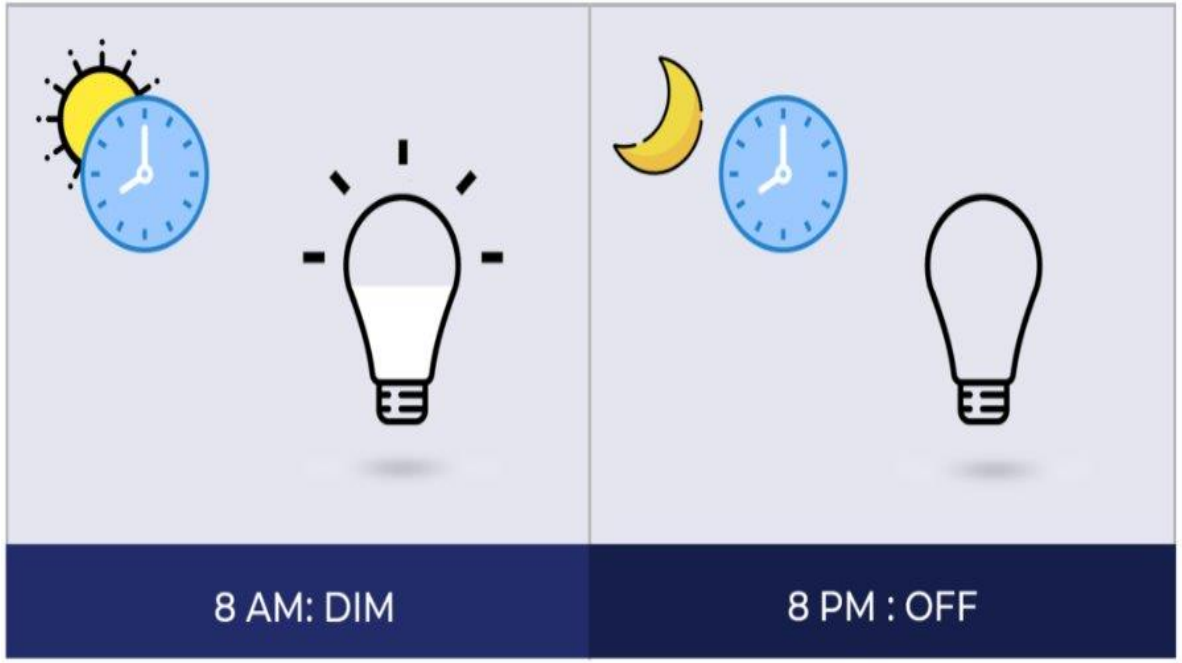
Overview

- Facility lighting is needed to provide visibility for building occupants, aesthetic atmosphere for spaces or rooms and for life safety.
- Lighting control systems provide lighting for occupants of the building as needed in an efficient manner, consistent with any applicable building and energy codes.
- The need for lighting in a building varies by the type of building, spaces within the building, time of day, and occupancy of the building.

LIGHTING CONTROL STRATEGIES

Scheduling

A control system may
have a predetermined schedule when
lights are turned on and turned off.

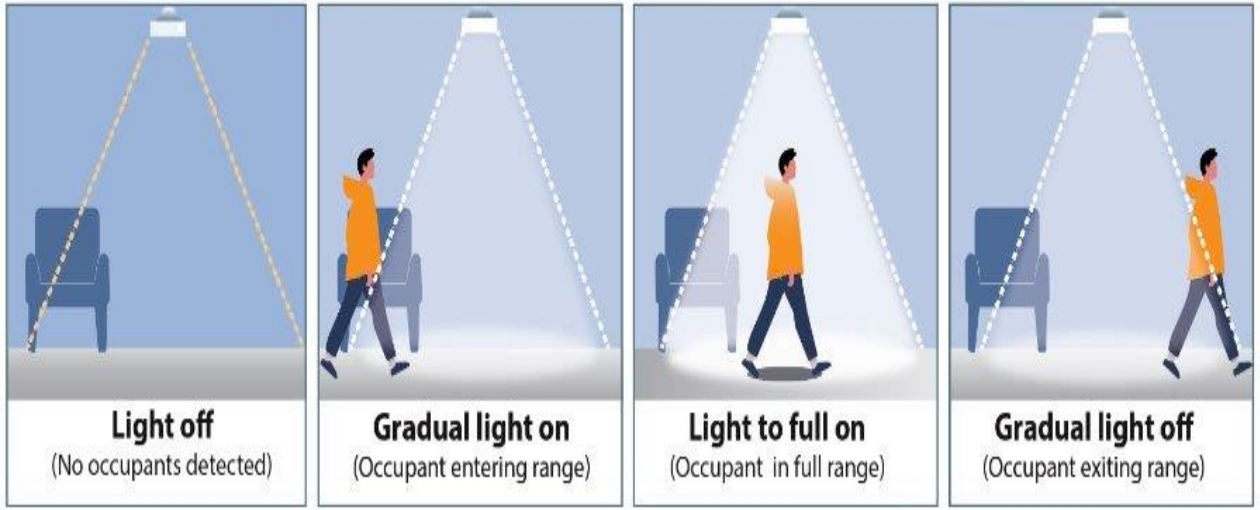


https://wisilica.com/company/lighting-control-strategies-offer-better-control-and-energy-savings/

LIGHTING CONTROL STRATEGIES

Occupancy sensors

For spaces in a building where occupancy is difficult to predict (such as meeting rooms or restrooms), lights may be turned on and off based on a lighting control system device sensing occupancy.



https://www.asmag.com/rankings/m/content.aspx?id=30803

LIGHTING CONTROL STRATEGIES

Daylighting

To reduce the need and cost of lighting spaces a control system utilizes natural light as much as possible. This is sometimes called “daylight harvesting” or “daylighting.”

The lamp turns on at full or dims to maintain the lux level, light output regulates accroding to the level of natural light available.

When natural light is sufficient, light dims to minimum but never switches off.

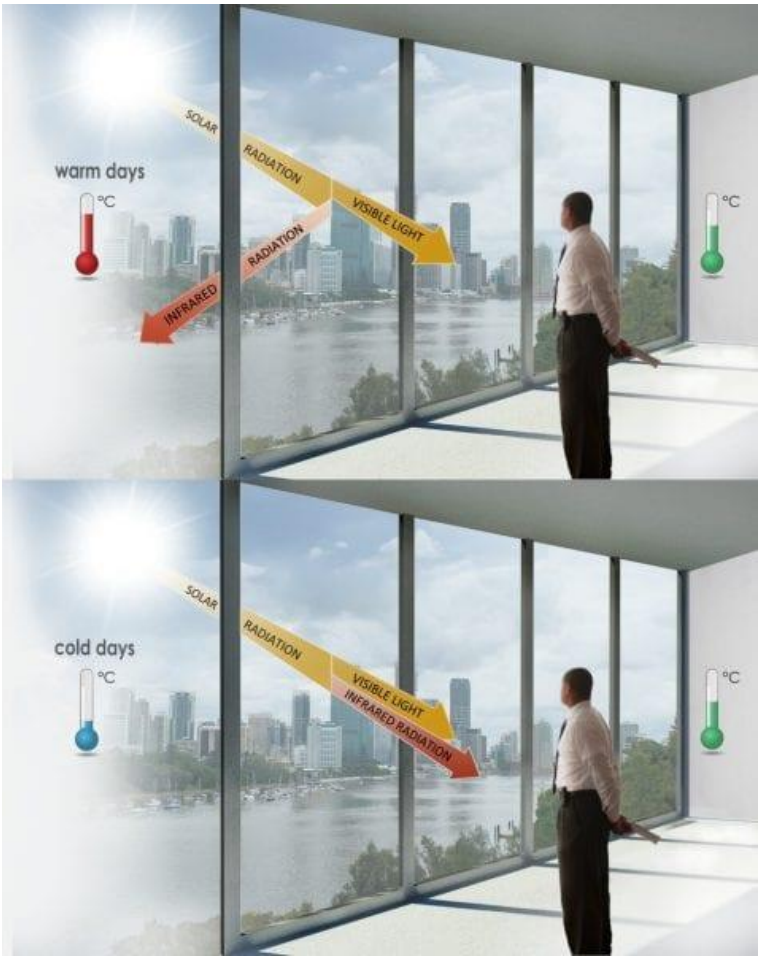
Switch off light manually while needed.

https://www.agcled.com/blog/save-energy-dali-v-daylight-sensors.html

LIGHTING CONTROL STRATEGIES

Window coatings

Schematic diagram of an ideal smart window reflecting infrared radiations in warm days and allowing it to enter in cold days (bottom), while remaining transparent in visible region in both climate conditions.



https://www.advancedsciencenews.com/infrared-regulating-smart-windows-based-organic-materials/

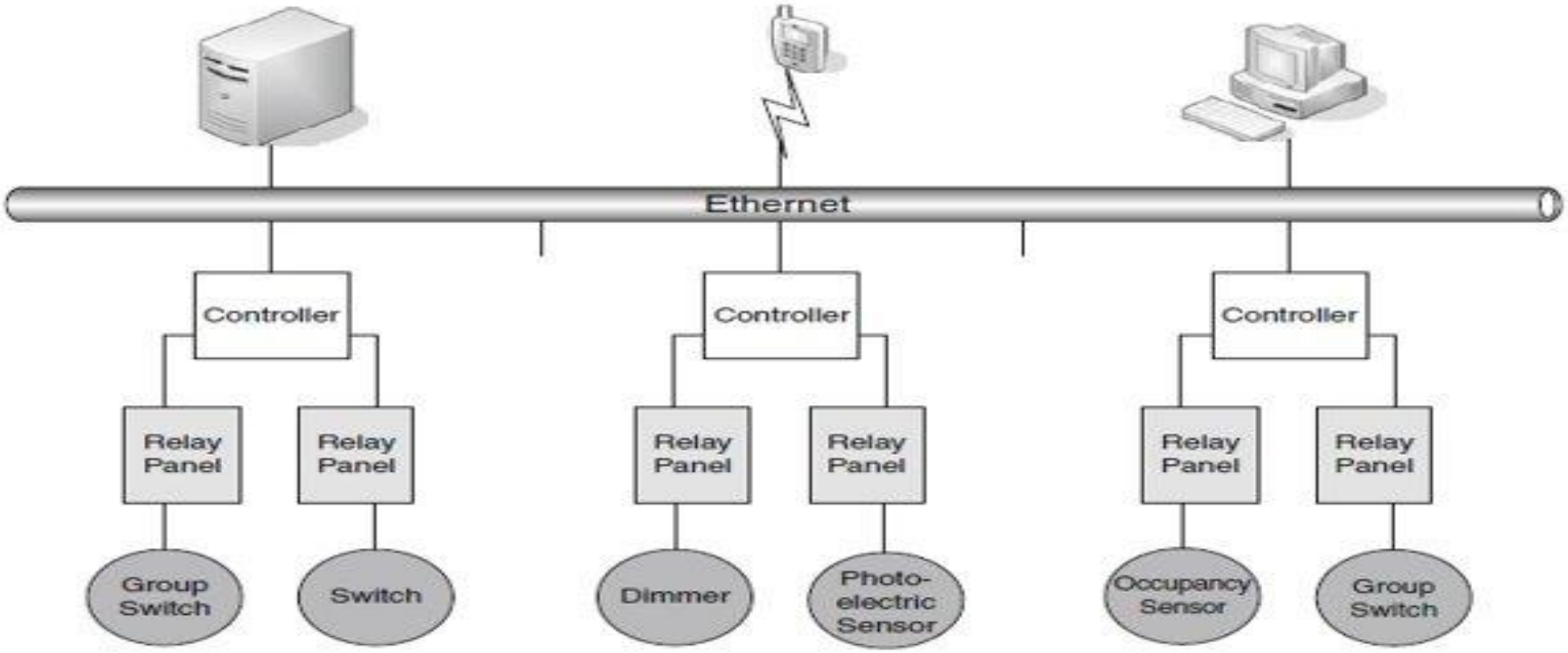
LIGHTING CONTROL SYSTEM

- The heart of the lighting control center is typically a server that is web-enabled and interconnected to other facility technology systems, a workstation with a GUI interface and client software for system administration.
- The networked system allows any authorized individual, including tenants or other occupants, to adjust their lighting through the network or a web browser.

LIGHTING CONTROL SYSTEM

- One approach to the lighting control system is the use of intelligent controllers.These controllers are distributed throughout a facility and manage downstream relay panels.
- The controllers and the system server are networked via an Ethernet network, usually sharing schedules and overrides. The controller may have a user interface panel which can be used instead of a system workstation to program and monitor the lighting control system.

LIGHTING CONTROL SYSTEM



A Typical Light controlling system

Source: James Sinopli, Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

RELAY PANELS

- Relays are switches that open and close circuits electromechanically or electronically.
- Relay panels are typically mounted next to the electrical circuit breaker panels.
- The circuit breaker panel feeds into the relay panel with the relays in the relay panel acting as a switching device for the circuit.
- Each relay can be individually programmed through the system controller or the relay panel.

DIMMERS

- Dimmer modules manage low-voltage switch and line voltage output controls of the dimmer's lighting loads.
- Stand-alone dimmers typically have status indicators, analog inputs for photo-cell or occupancy sensors, diagnostics, and are able to optimize responses for various types of lighting fixtures.
- Dimmers can be used for specific spaces such as areas with audio visual presentations or throughout the total system for managing large facilities.

Source: https://www.ubuy.co.in/product/2B181IIO-led-dimmer-dc-12v-24v-lighting-dimming-controller-30a-12-volt-24-volt-light-dim-switch-easy-solution

INTEGRATION INTO BUILDING AUTOMATION SYSTEMS

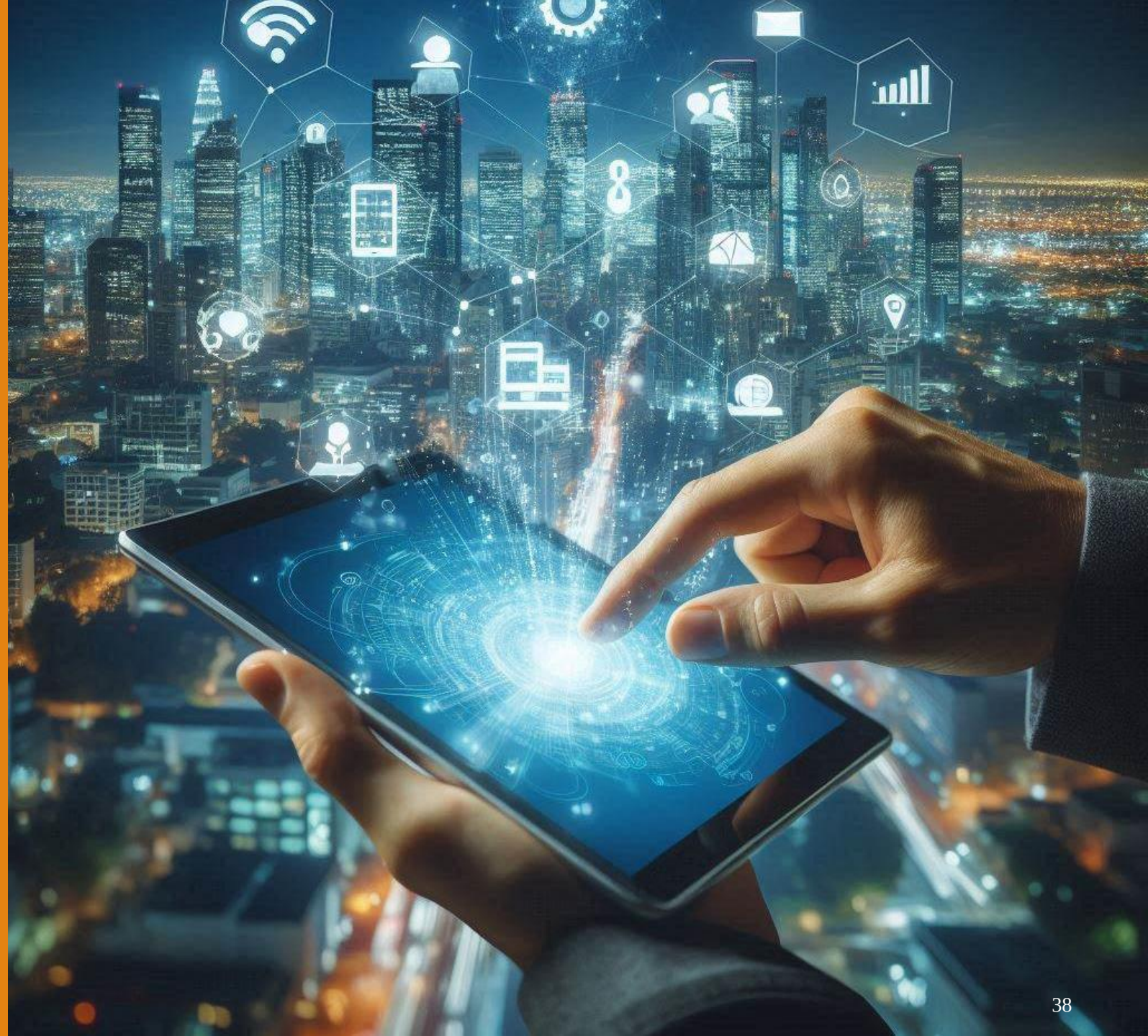
- Lighting systems provide a life safety function, assisting in security or lighting evacuation pathways from a building.
- Lighting systems may be integrated with fire alarm systems, security systems or emergency power generators.
- In the case of a fire alarm or loss of normal power the lighting control system may turn on key emergency lighting fixtures.
- Data and information from a lighting system are also an integral part of an overall energy strategy and at a given facility or business level, need to be considered with HVAC systems, metering and building plug loads.

SUMMARY

- Overview
- Control strategies
- Lighting system control
- Typical light controlling system
- Components
- Integration into building automation system

https://www.youtube.com/watch?v=4B6kQrxM5-4
https://www.youtube.com/watch?v=yZil-n5ppQw
https://www.youtube.com/watch?v=xnMySo_Nwi8

LECTURE – 10
ACCESS CONTROL SYSTEM



ACCESS CONTROL SYSTEM

Overview:

- Manual control systems rely on human actions, while access control systems (ACS) use electronic actions to restrict access to a resource. The vital importance of access control systems has increased along with security for buildings.

- The basic or typical building access control system operates so that a person presents a card to a card reader for a particular door and based on the information on the card and the system parameters for that person, door, and facility, the system either unlocks the door to allow the person to pass through or refuses entry.

WORKING OF ACCESS CONTROL SYSTEM

- When a person arrives at a door and presents an access card with encoded credential information to a card reader that information is then passed to the local control panel. In a centralized database system, the control panel passes the information on to the host computer with additional information regarding the location of the door and the time the card was presented.
- The host computer then verifies the information and compares the access level to the door location and the time of day.
- Upon verification, the host computer sends a command to open the door to the control panel and the person is allowed access to the building

COMPONENTS OF ACS

1. A central host computer or server.

2. Control panels or system controllers connected to the host computer.

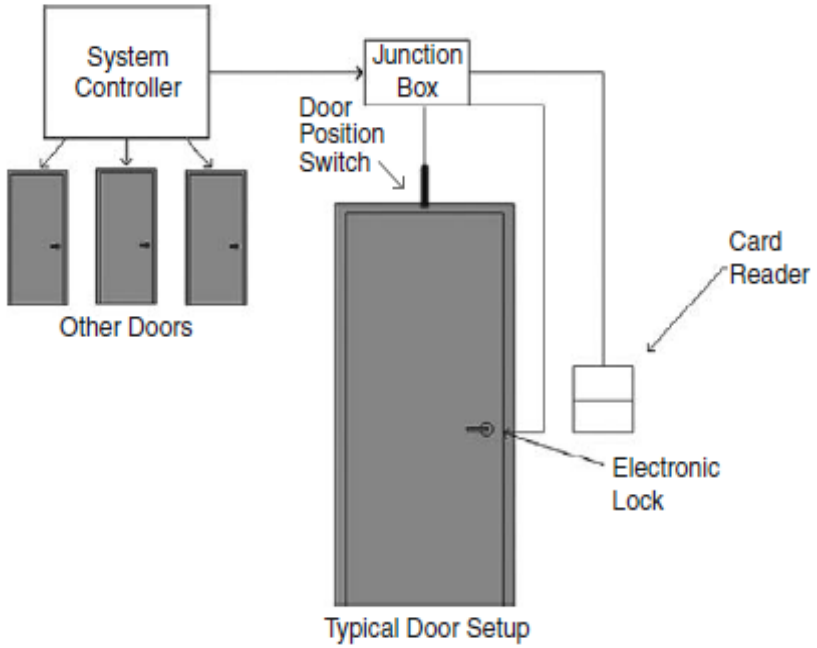
3. Peripheral devices such as card readers, door contacts, sirens, and sensors connected to the control panels.

COMPONENTS OF ACS

A central host computer or server:

The server or host computer houses the operating parameters and the database for the access control system.

The host computer is networked and communicates with the control panels which collect data on events and alarms from peripheral devices.



Access control deployment at a typical door

Source: James Sinopli, Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

COMPONENTS OF ACS

Control panels:

- Control panels are usually enclosed printed circuit boards with connections to all peripheral devices in their area. These peripheral devices may include door hardware (such as a card reader, door position switch, or door strike) and other inputs and relays as required.
- The control panel manages the peripheral devices and communicates between the host computer and the peripheral devices.

COMPONENTS OF ACS

Functions the control panels :

- Consolidation of all connections to peripheral devices.
- Provision of power, as needed, to peripheral devices.
- Management of peripherals when communications to the host computer are absent or when acting in a distributed manner.

COMPONENTS OF ACS

Peripheral Devices:

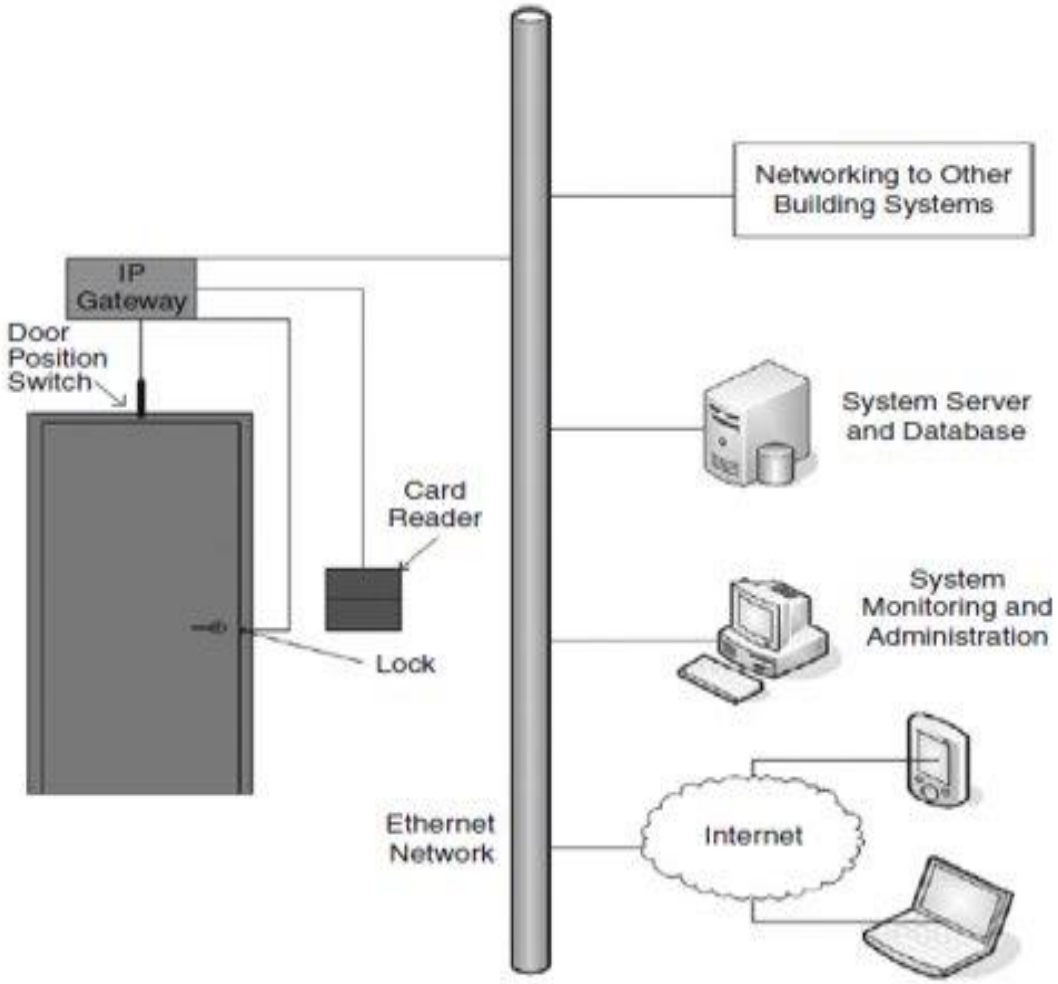
- Door contacts
- Request to exit
- Electrified door hardware
- Card readers

CURRENT TRENDS IN ACCESS CONTROL

- Access control systems that incorporate IP and power-over-Ethernet (POE) components have been introduced to the marketplace in recent years.
- The evolution of access control systems to structured cabling, IP protocols, and POE was inevitable; a similar evolution took place in telephone and video surveillance systems.
- This transformation leverages the existing IT infrastructure, eliminates the need for local power, consolidates and saves labor costs for cable installation, reduces the time to install system devices, is more scalable and provides a large base of management tools and support. The move is subtly but surely changing the design and deployment of access control systems.

CURRENT TRENDS IN ACCESS CONTROL

IP access control:



Source: James Sinopli, Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

BENEFITS OF POE

Cost-saving:

- Provides significant infrastructure cost benefits.
- Does not need the junction box above the door, as well as the high-voltage power supply, network or building controllers, and local battery packs.
- This infrastructure leverages the existing IT network system, which helps to reduce the cost.

BENEFITS OF POE

Flexible installation:

- The PoE cables transmit both power and data together to each device attached to the local area network (LAN).
- Normally, a device must be close to an AC power outlet for optimal performance.
- PoE expands the flexibility of device positioning, allowing devices to be installed easily in a network without the concerns of their proximity to AC outlets.
- With a PoE access control system, there is no need to spend time installing AC outlets since a PoE switch can provide power for the devices, saving time and resources.

BENEFITS OF POE

Easy and safe management:

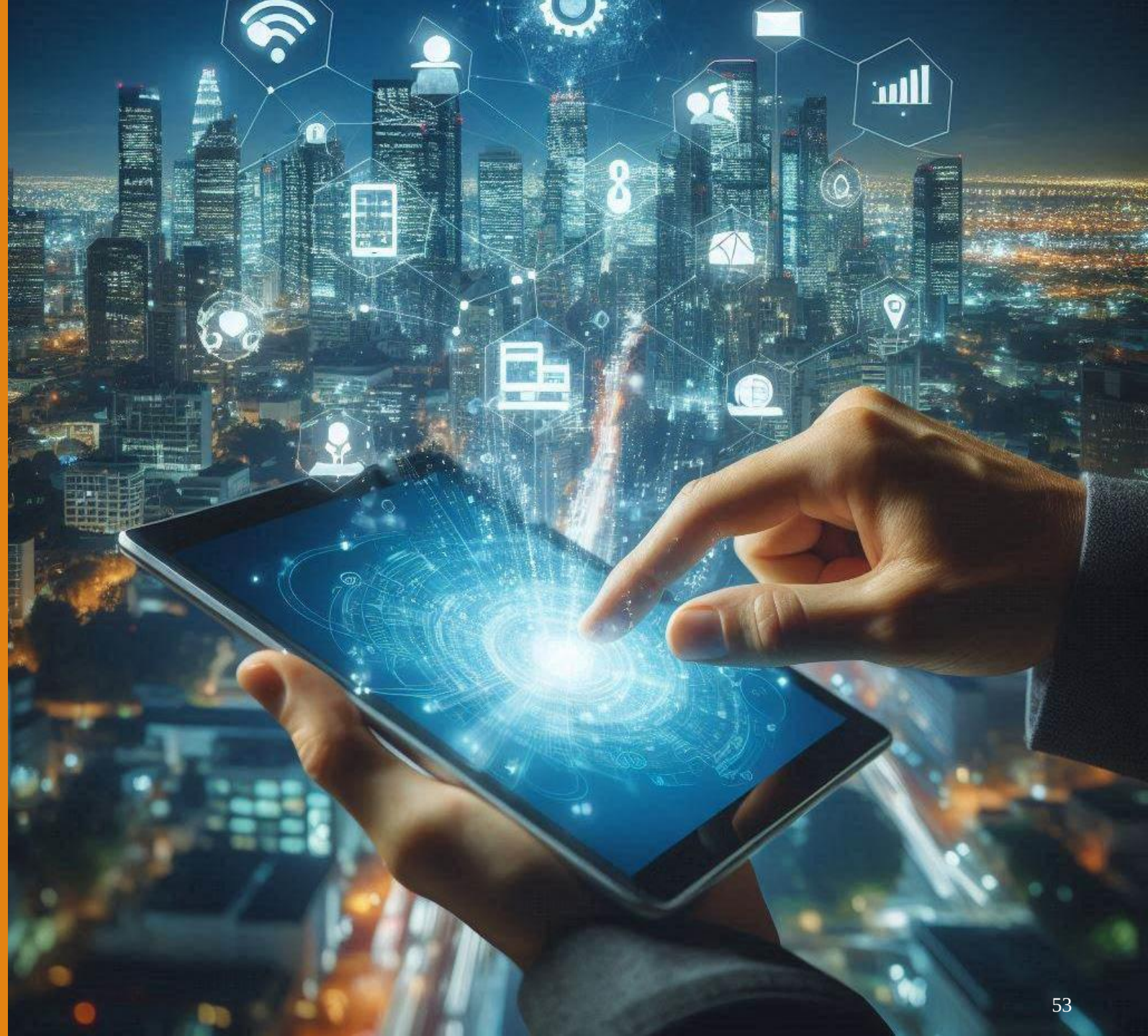
- The power over Ethernet system uses a centralized power source, allowing remotely powered devices to connect and disconnect during service reconfigurations and disruptions easily.
- PoE technology helps protect the whole network investment because it has established an 802.3 Ethernet protocol extension and is supported by various data rates that reach up to 10 Gbps.
- The PoE access control system broadens the security initiatives of building premises by supporting CCTV devices for better surveillance.
- Digital signals allow the cameras to monitor who is accessing the premises.

SUMMARY

- Overview
- Basic components of ACS
- Current trend in ACS
- IP access control and benefits

https://www.youtube.com/watch?v=2QTFiQVdrgg
https://www.youtube.com/watch?v=DeyruqSZnhE

LECTURE – 11
FIRE ALARM SYSTEM



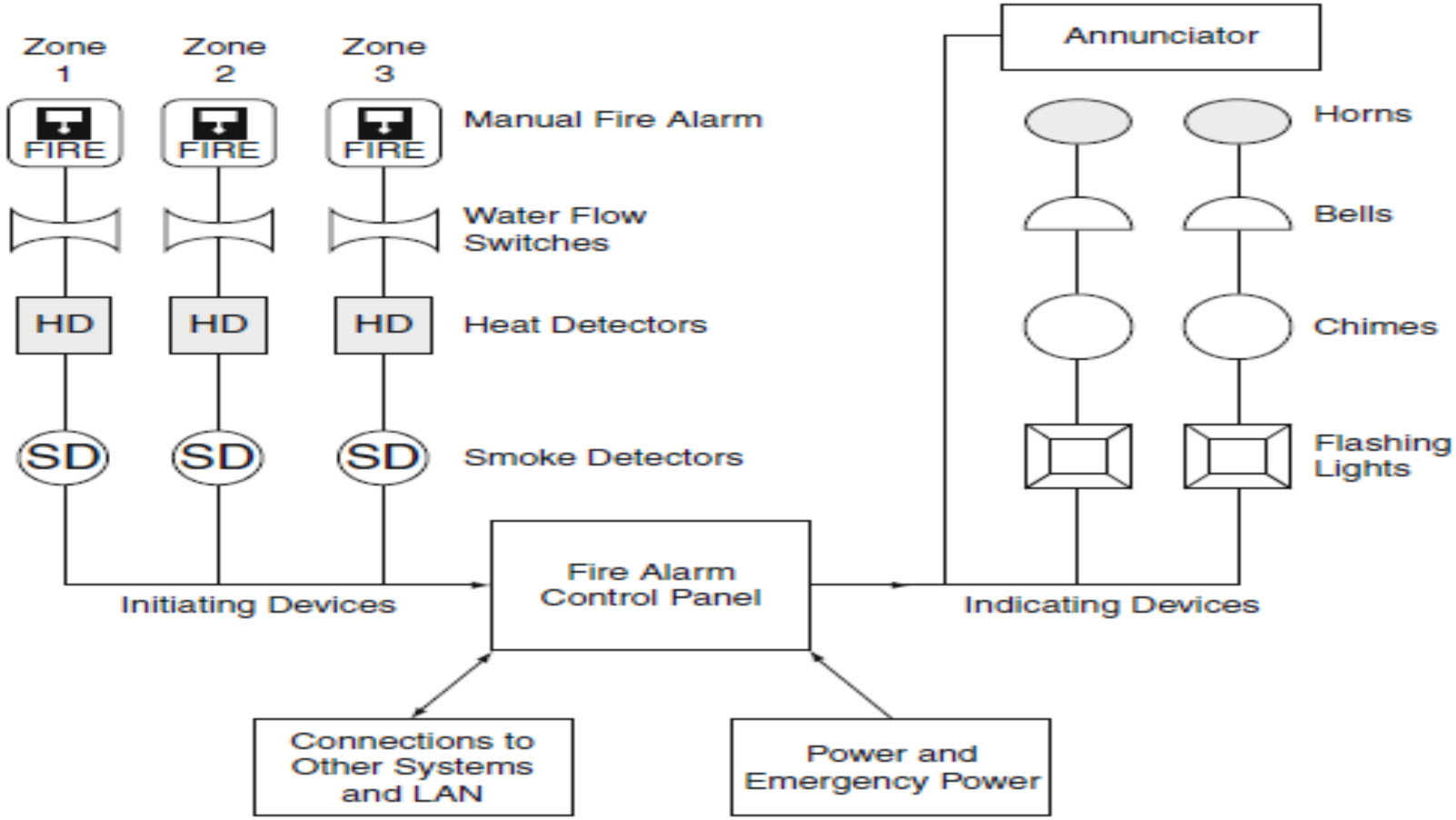
OVERVIEW

- Fire alarm systems are the primary life safety system for every building. Properly deployed, a fire alarm system reduces the probability of injury or loss of life and limits damage due to fire, smoke, heat and other factors.
- Because of their criticality, the codes, regulations and standards affecting the design and installation of fire alarm systems are wide ranging and detailed.
- Their design and installation must involve qualified, licensed, experienced professionals and more important, the coordination and approval of the local authority having jurisdiction (AHJ).
- A fire alarm system will initiate communications to other building automation and security systems to facilitate evacuation from the building and containment of the fire. Such systems include the following:

OVERVIEW

- Signaling the HVAC system to restrict and contain smoke, heat and fire through dampers and fans
- Using the access control system to clear a path for building evacuation by opening doors, unlocking secured doors, and releasing powered exterior doors
- Using the access control system to contain and prevent the spread of fire and smoke by closing interior doors
- Triggering emergency power for the fire alarm system and related systems operation, exit signs, and lighting for building exit routes
- “Capturing” the elevator and shutting down its operation

BASIC FIRE ALARM SYSTEM



MAJOR COMPONENTS OF FIRE ALARM SYSTEM

Fire Alarm Control Panel

- The heart of a fire alarm system is the primary fire alarm control panel (FACP). The FACP monitors system integrity and starts all sequences of operation for system detection, suppression, and notification. FACPs are typically microprocessor based with software for communications, processing and decision making.

MAJOR COMPONENTS OF FIRE ALARM SYSTEM

Annunciator panel:

A major component to the system is the annunciator panel attached to the FACP. The annunciator provides visual and audible indications that an alarm has been initiated as well as the location of the alarm. It may also identify the functions that could affect the fire and the building occupants in the area.

MAJOR COMPONENTS OF FIRE ALARM SYSTEM

Fire detection systems:

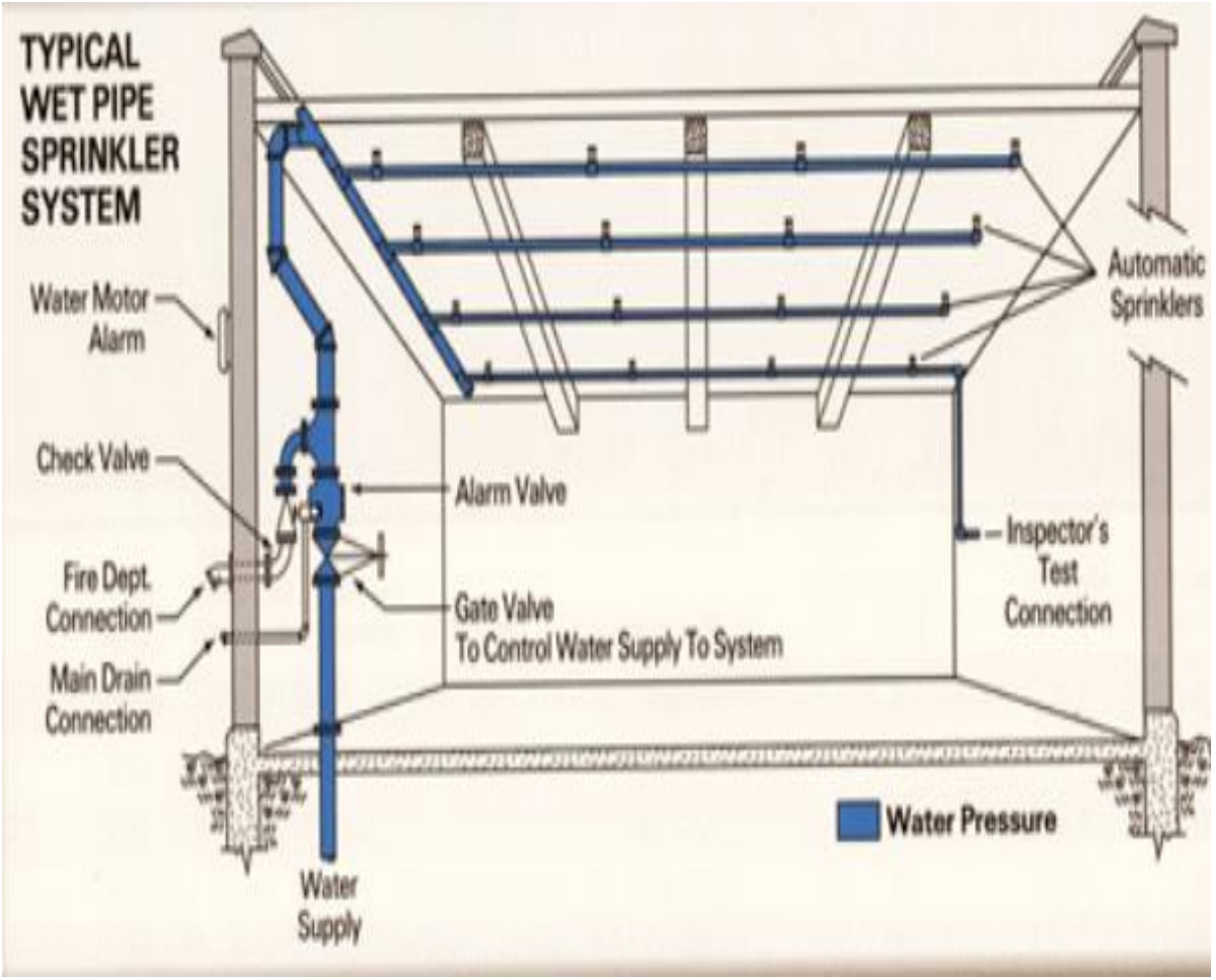
Fire consists of smoke, heat, and light. The system components that detect the fire and initiate an alarm monitor one or more of the fire's characteristics. The detection components of a fire alarm system are typically located in ceilings, HVAC ducts, mechanical and electrical areas and equipment rooms. In most buildings, two types of fire detectors are installed: smoke detectors and heat/rise detectors.

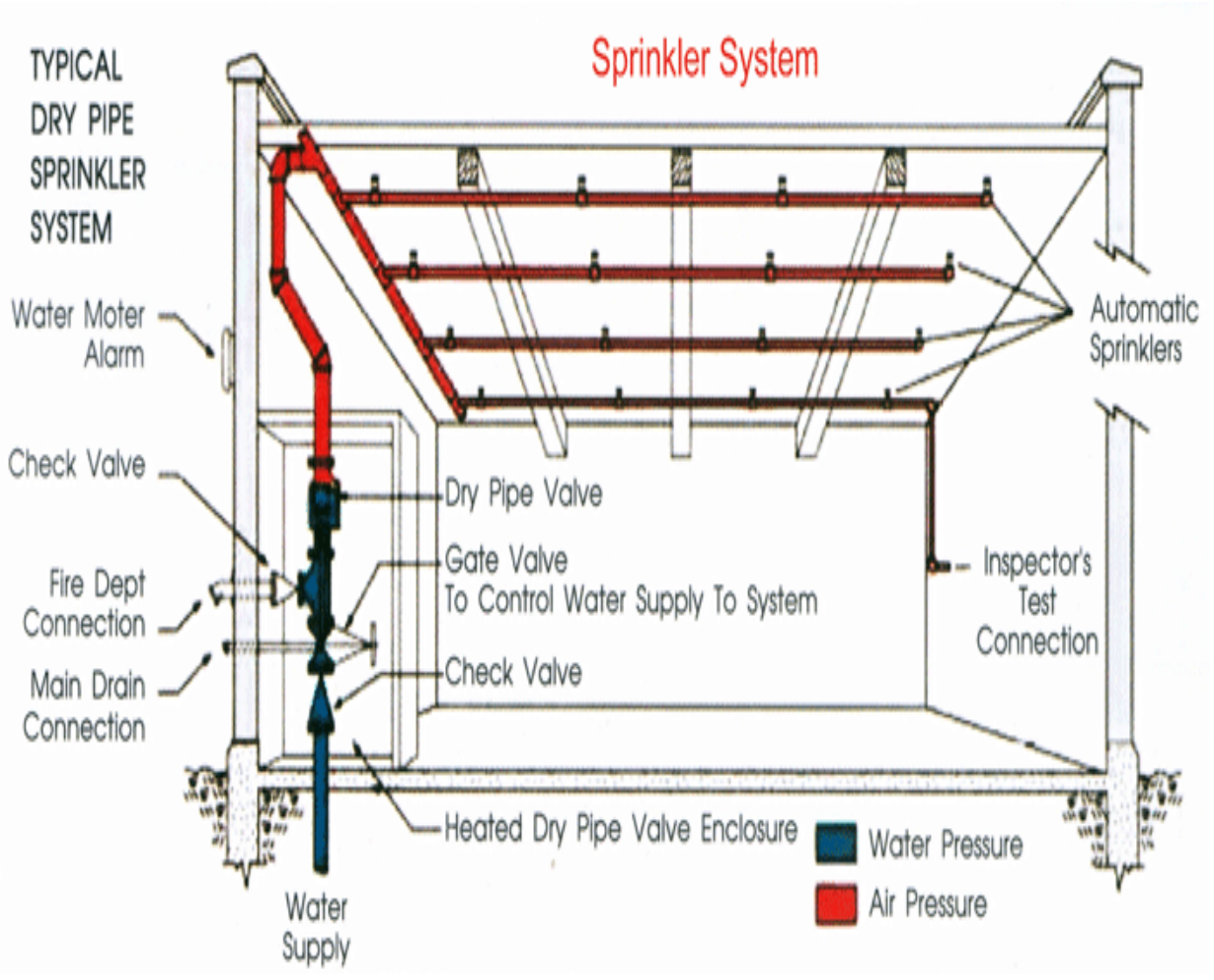
Source: https://scotlandhomesafe.co.uk/blogs/news/the-difference-between-a-smoke-heat-alarm

MAJOR COMPONENTS OF FIRE ALARM SYSTEM

Fire suppression systems:

- Fire suppression systems include the following:
- Wet sprinkler systems, which may comprise various switches and flow detection equipment that is monitored and managed
- Dry sprinkler systems, which may include pressure switches that are
- monitored and managed
- Fire suppression systems may also include monitoring and supervision equipment


TYPICAL
WET PIPE
SPRINKLER
SYSTEM
Water Motor
Alarm
Check Valve
Fire Dept.
Connection
Main Drain
Connection
Alarm Valve
Gate Valve
To Control Water Supply To System
Automatic
Sprinklers
Inspector's
Test
Connection
Water Pressure
Water
Supply


Sprinkler System
TYPICAL
DRY PIPE
SPRINKLER
SYSTEM
Water Moter
Alarm
Check Valve
Fire Dept
Connection
Main Drain
Connection
Dry Pipe Valve
Gate Valve
To Control Water Supply To System
Check Valve
Automatic
Sprinklers
Inspector's
Test
Connection
Heated Dry Pipe Valve Enclosure
Water Pressure
Air Pressure
Water
Supply

MAJOR COMPONENTS OF FIRE ALARM SYSTEM

Notification devices:

Once a fire is detected a building's occupants must be notified to evacuate the building. Fire notification devices use audio, visual, or a combination signaling to notify occupants. These devices are typically DC-powered so that they can operate on backup batteries.

Fire alarm notification devices include, but are not limited to, the following:

- Bells
- Chimes
- Horns
- Speakers
- Strobes, including strobe lights combined with other devices

MAJOR COMPONENTS OF FIRE ALARM SYSTEM

Mass notification systems:

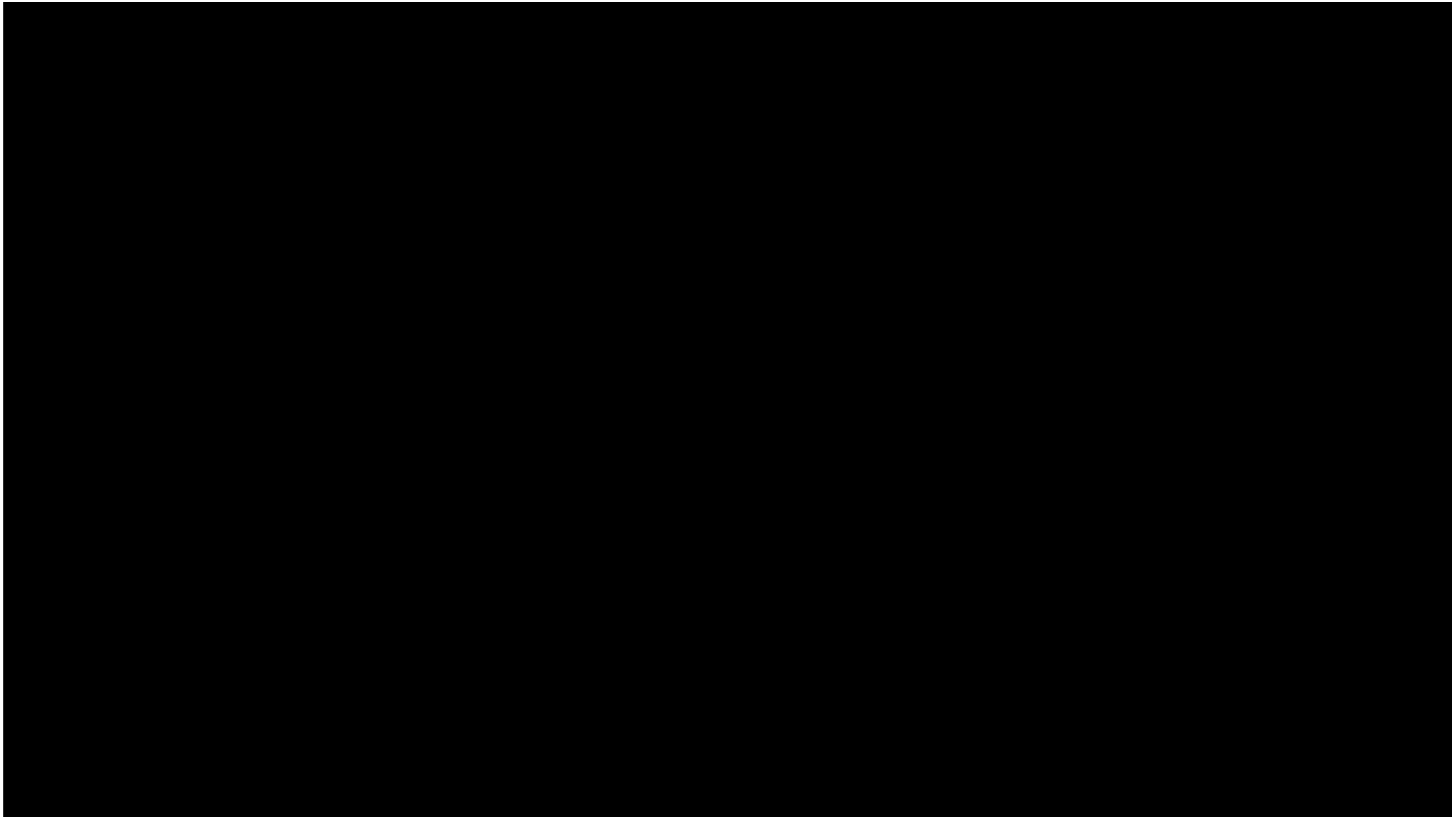
- Mass notification systems (MNS) are used to provide real-time descriptive information and directions to people during fire and non fire emergencies.
- For example, many fire alarm systems have a paging system to notify building
- occupants of a fire situation en masse.
- Mass notification systems cover not only buildings, but campuses, cities, regions, and the globe; thus, MNS types are for buildings, wide-area notification, and distributed notification. MNS are used for fire situations, public alerts, emergency situation conditions and warnings such as for severe weather.

STANDARDS FOR FIRE ALARM SYSTEMS

- National Fire Protection Association (NFPA)
- Underwriters Laboratories (UL). NFPA 70, 72, and 101 address the National Electrical Code, National Fire Alarm Code, and Life Safety Code, respectively.
- Indian code IS 15908 : 2011 deals with selection, installation and maintenance of control and indicating equipments for fire detection and alarm system.

SUMMARY

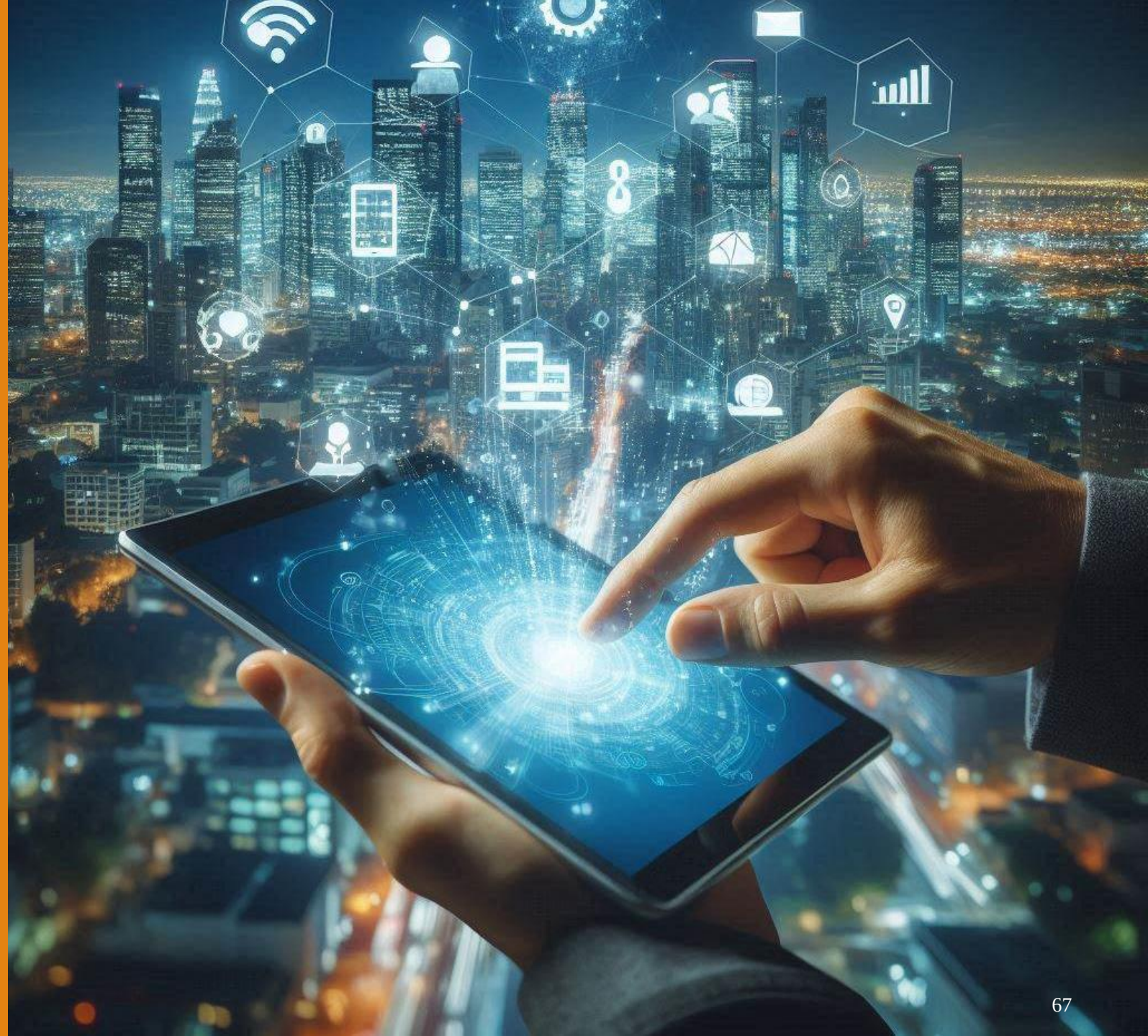
- Basics of fire alarm systems
- Various components of fire alarm systems
- Codes related to fire alarm systems



Source: https://www.youtube.com/watch?v=cVjyDgFrb2g

LECTURE - 12

VIDEO IPTV



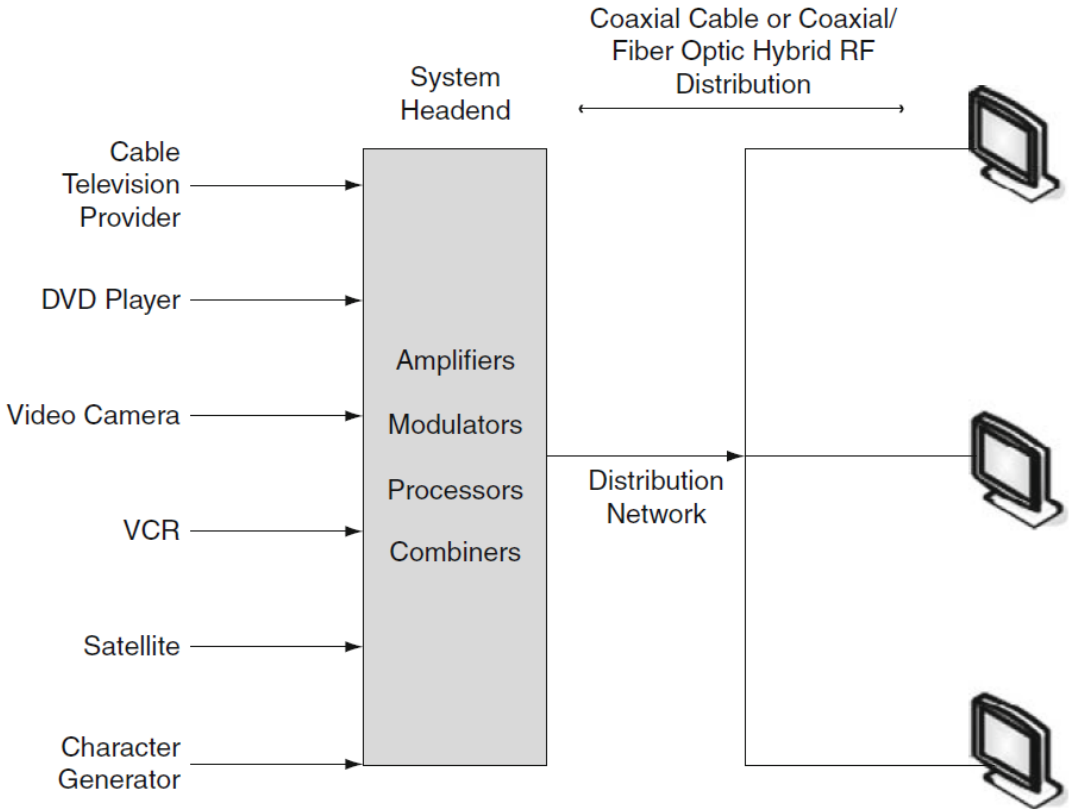
VIDEO IPTV

Overview

- IPTV, or Internet Protocol Television, is a technology that delivers television/video/audio/text/graphics/data over the internet rather than through traditional broadcasting methods like satellite or cable.
- The key differences between IPTV and traditional TV broadcasting is the delivery mechanism and the range of features and services available with IPTV.

TRADITIONAL VIDEO DISTRIBUTION

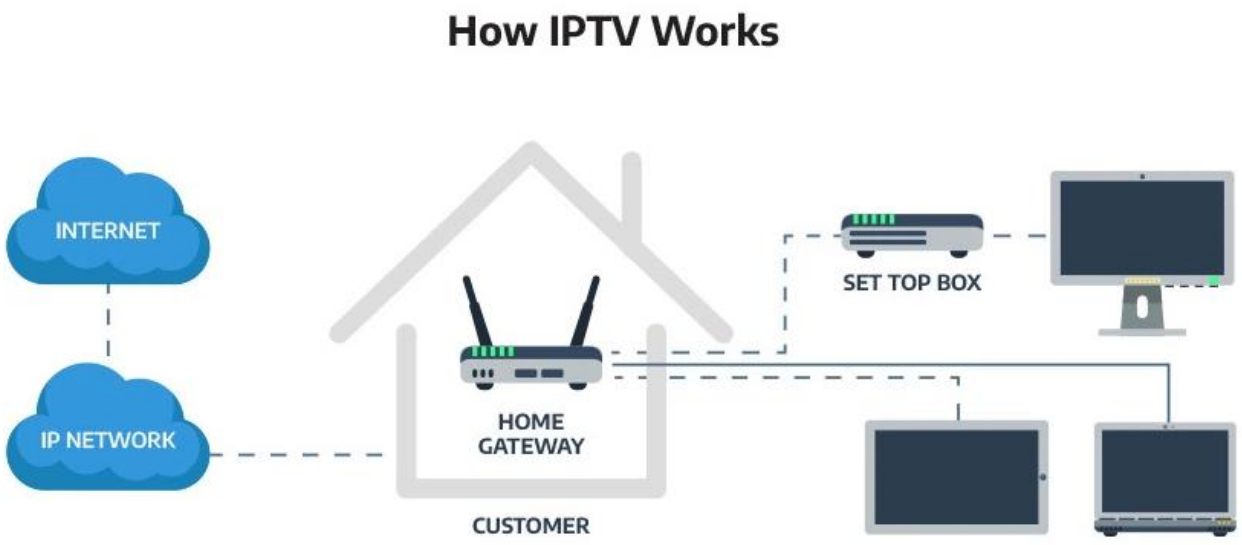
- Traditional TV broadcasting relies on sending television signals through terrestrial, satellite, or cable networks. These signals are received by antennas, satellite dishes, or cable connections.



Refer: Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

IP TELEVISION -VIDEO DISTRIBUTION

IPTV, utilizes internet protocols to deliver television content. The TV signals are encoded into IP packets and transmitted over IP-based networks, such as broadband internet connections.



Source: https://www.techtarget.com/searchnetworking/definition/IPTV-Internet-Protocol-television

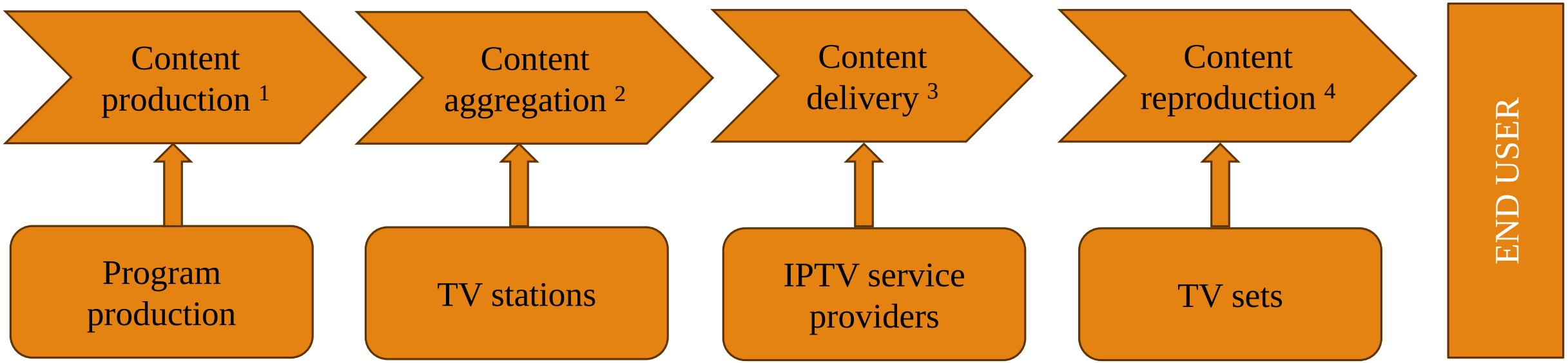
IPTV AND TRADITIONAL V/S TV BROADCASTING

Traditional TV broadcasting	IPTV
Relies on sending television signals through terrestrial, satellite, or cable networks	Utilizes internet protocols to deliver television content
Requires a dedicated infrastructure of broadcast towers, satellites, or cable networks to transmit signals to a wide audience	Leverages existing internet infrastructure, enabling content delivery over broadband networks.

IPTV AND TRADITIONAL V/S TV BROADCASTING

Traditional TV broadcasting	IPTV
Face limitations in reaching remote or rural areas due to infrastructure constraints.	Can reach a wider audience since it relies on internet connectivity, which is more readily available in urban, suburban, and rural areas with broadband access.
Delivers a fixed set of channels to all viewers simultaneously	Users can access a wider range of content, including live TV channels, video-on-demand (VOD), and recorded shows.

IPTV NETWORK ELEMENTS

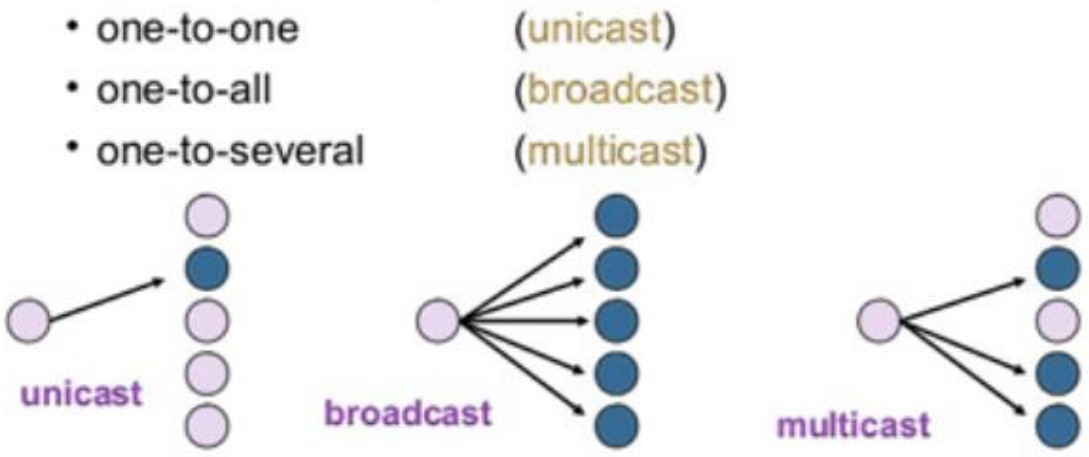


1. *Content sources are where television channels, video-on-demand (VOD) libraries, and other media content originate.*
2. *Content aggregation and management systems gather, organize, and manage the vast array of content available for distribution.*
3. *It consists of servers located strategically in different geographical locations to reduce latency and ensure high-quality streaming.*
4. *To convert the content into a format suitable for processing the electronic device the end user*

Source: https://www.researchgate.net/publication/299462199

VIDEO TRANSMISSION – DELIVERY MODE

Delivery of video over a standard data network can be performed in several modes: *programming can be sent to just one device (unicast), to all open devices at the same time (broadcast) and to multiple devices at the same time (multicast)as shown in figure.*



Source: https://www.techtarget.com/searchnetworking/definition/IPTV-Internet-Protocol-television

VIDEO TRANSMISSION – DELIVERY MODE

1. <u>Unicast:</u>

- Each viewer receives a separate stream of video content from the server.
- Unicast delivery is suitable for on-demand content and ensures personalized viewing experiences.
- It requires significant bandwidth and server resources, making it less scalable for large-scale live streaming events with a high number of concurrent viewers.

.

VIDEO TRANSMISSION – DELIVERY MODE

2. Broadcast:

- Involves sending video content to all devices on a network simultaneously, without the need for individual requests from users.
- This is commonly used in traditional TV broadcasting, where television signals are transmitted over the airwaves or through cable networks to all viewers within a geographic area.

VIDEO TRANSMISSION – DELIVERY MODE

3. Multicast:

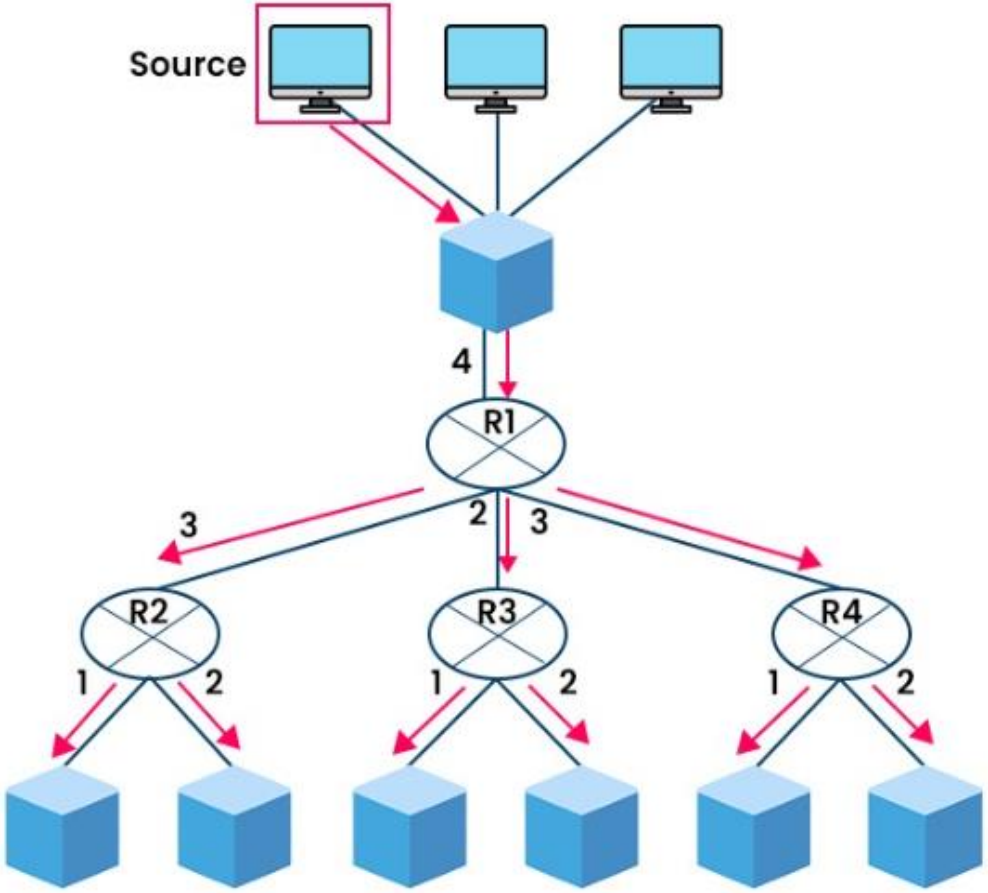
- It enables efficient delivery of video content to multiple recipients simultaneously.
- Instead of sending individual streams to each viewer, multicast technology replicates the video stream at network nodes, allowing multiple users to access the same content stream.
- This conserves bandwidth and reduces network congestion, making it ideal for live broadcasts, such as sports events or news updates.

BROADCAST V/S MULTICAST

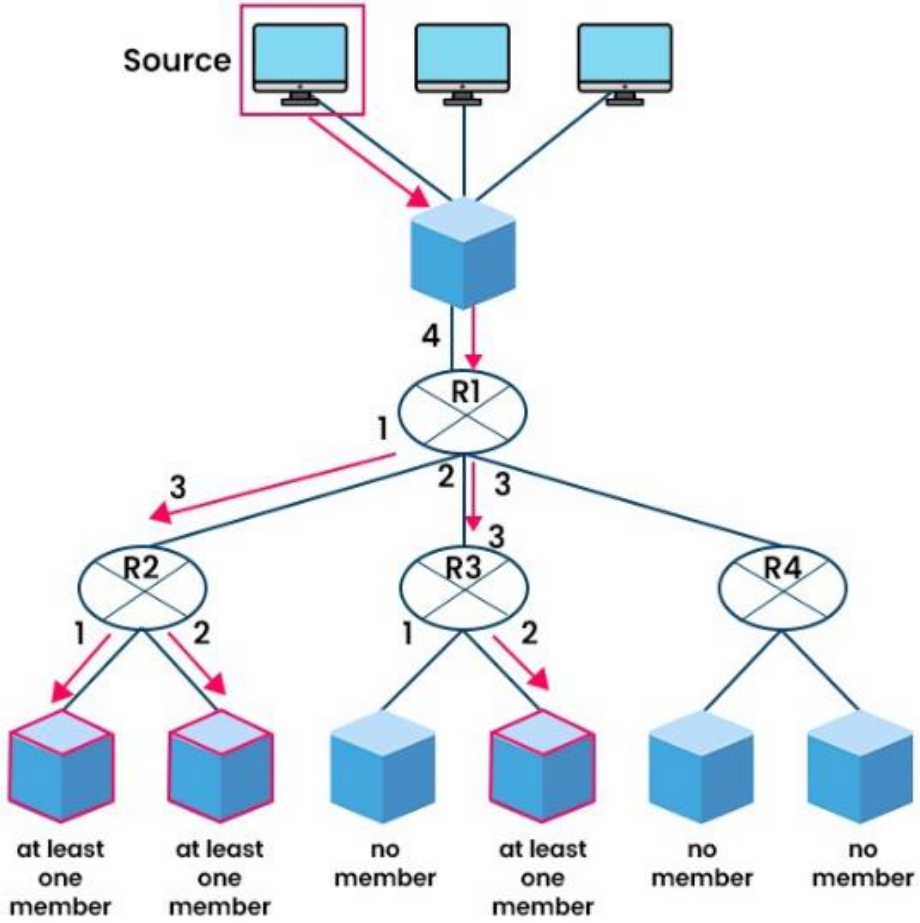
Factors	Broadcast	Multicast
Basic	One-to-all	One-to-many
Transmission	The sender transmits data to all receivers in the network, regardless of their interest.	The sender transmits data to a group of receivers who have joined the multicast group.
Management	No group management is required.	Group management protocols are needed to join and leave multicast groups.
Bandwidth	Wastes bandwidth as data is sent to uninterested receivers.	Saves bandwidth as data is sent only to interested receivers.
Traffic	Generates more network traffic as data is replicated at every router.	Generates less network traffic as data is replicated only at branching points.
Process	Simple and easy to implement.	Complex and requires more processing power.
Security	Less secure as data is exposed to all receivers.	More secure as data is delivered only to authorized receivers.

Source: https://www.pynetlabs.com/difference-between-broadcast-and-multicast/

BROADCAST V/S MULTICAST



BROADCAST



MULTICAST

Source: https://www.pynetlabs.com/difference-between-broadcast-and-multicast/

APPLICATIONS of IPTV

- *Media Retrieval*: video programming stored in digital or analog format can schedule, or "call up," in real time.
- *Video conference*: Video conferencing technology addresses the cost and potential inefficiencies of live meetings.
- *Distance learning:* The application may also be augmented with graphics transmissions.
- *Live feeds from camera*: A platform that allows users to access live and on-demand content.

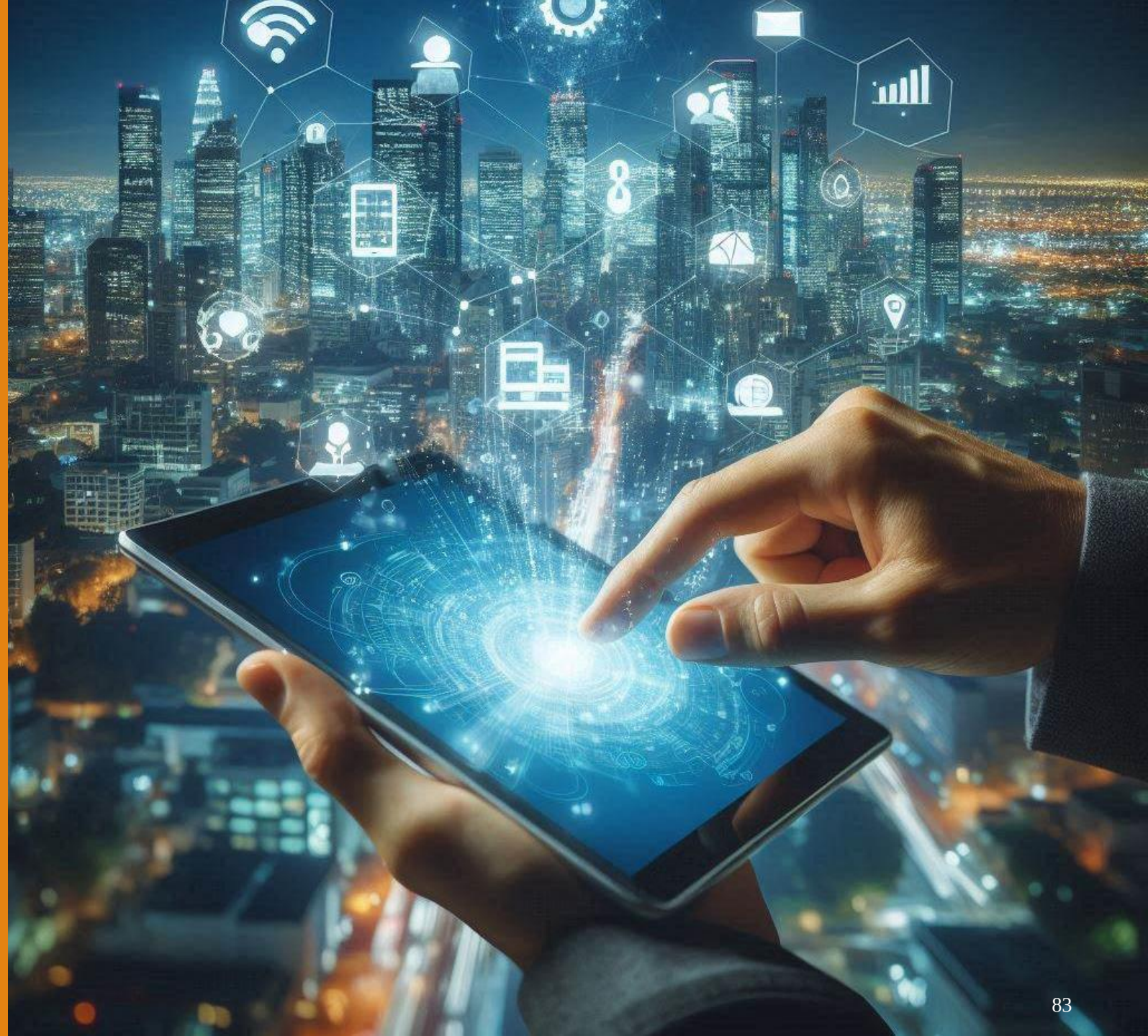
SUMMARY

- IPTV represents a modern approach to delivering television content, leveraging the internet to provide a flexible, interactive, and personalized viewing experience.
- Its scalable delivery, efficient use of network infrastructure, and potential for cost savings, make it a compelling choice for both service providers and consumers.

https://youtu.be/64tyAvWBjic

LECTURE - 13

Audio visual system



AUDIO VISUAL SYSTEM

Overview

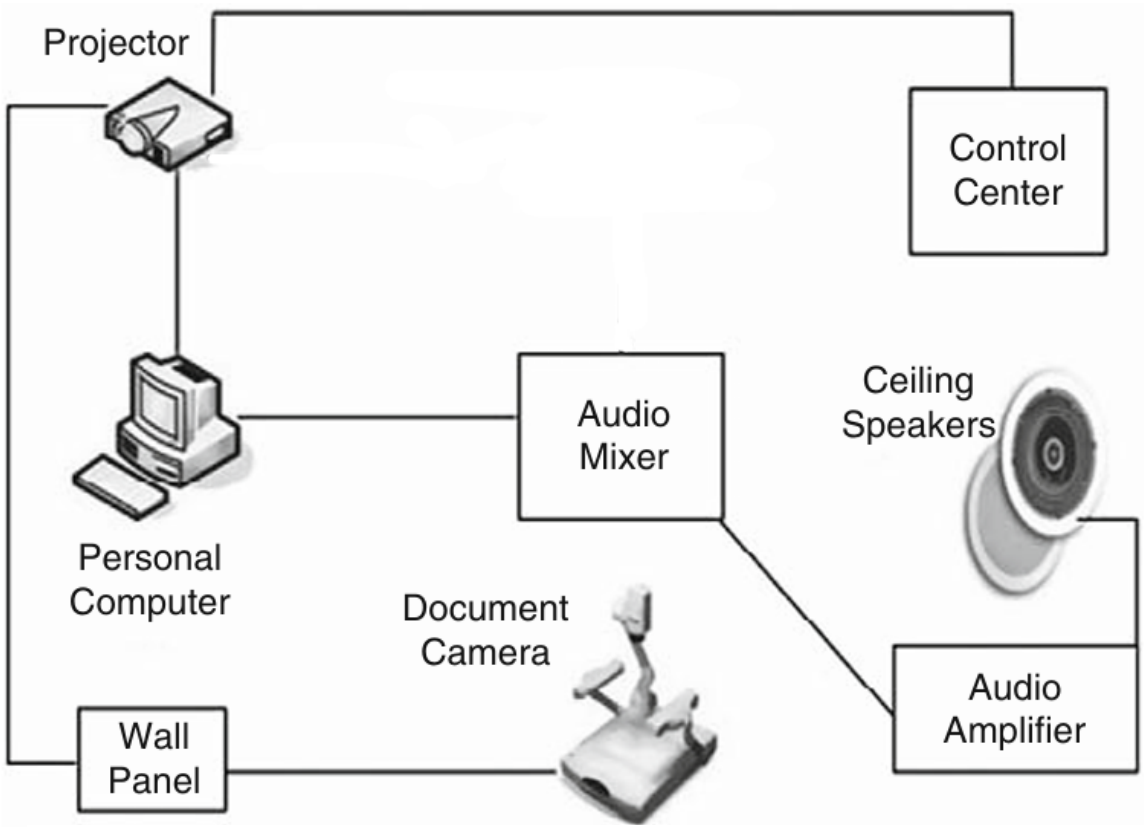
- Audiovisual (AV) systems encompass a wide range of technologies and equipment designed to enhance communication through audio and visual components.
- These systems are integral in various settings, including businesses, educational institutions, entertainment venues, and residential spaces.

AUDIO VISUAL SYSTEM

- AV systems play a vital role in enhancing communication, engagement, and productivity across diverse environments, leveraging a combination of audio, video, and control technologies tailored to specific applications and user requirements.
- Key components and considerations of AV systems are Display Devices, Audio Equipment, Control Systems, Signal Processing, Connectivity Infrastructure, Collaboration Tools, Control Room and Command Centers, Integration and Installation.

TYPICAL CLASSROOM AUDIO VISUAL SYSTEM

Audio visual systems are designed for the specific needs of certain rooms and spaces within a facility, such as meeting rooms or classrooms



Refer: Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

CORE COMPONENTS OF AUDIO VISUAL SYSTEMS

- Audio and visual sources
- Processing and management
- Destinations (speakers and displays)
- System control

1. AUDIO AND VISUAL SOURCES

- The sources of audio are generally from microphones, electronic instruments or programmed sources. Microphones convert acoustic energy (such as talking and singing) into electrical energy.

- The visual part of an audio visual system also has several sources such as cameras, broadcast/cable/satellite, document cameras, personal computers, data networks.

2. AUDIO AND VIDEO PROCESSING

- An equalizer processes the audio signal by increasing or decreasing the low, mid and high-level frequencies to provide the most pleasing audio sound possible.
- Video processing may involve amplifying or adjusting the timing, color, brightness, or contrast of the signal.
- Video processing equipment can include time base correctors to maintain the integrity of the signal and video processing amplifiers.

DESCRIPTIONS OF AUDIO PROCESSORS

Echo cancellation	Reduces the chance of sound output through a loudspeaker being reamplified into the sound input
Limiters	Limit the amplitude of the sound, restricting the volume to a predetermined setting.
Compressors	"Compress" the audio signal to a certain volume range avoiding both the loudest and the quietest sounds.
Expanders	Expand or increase the volume of the audio signal to a predetermined sound level.

Descriptions of Audio Processors

Gates	Gates block or eliminate audio sounds below a certain frequency level.
Automatic gain control	Automatically controls the volume of the signal at a specific frequency level.
Delay	Used in large venues where natural sounds can be heard after processed sounds and delay is used to synchronize the two.
Digital signal processing	Used with digital as opposed to analog audio and video signals.
Distribution amplifiers	Take a single audio input, amplify it, and distribute it to multiple audio outputs.

3. SPEAKERS AND DISPLAYS

Types of speakers

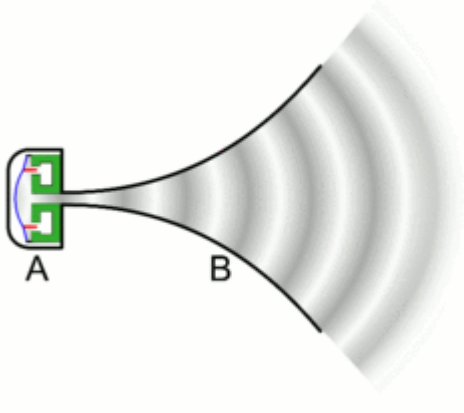
Loudspeakers

Cones

Compression drivers [1]

Used for reproduction of the bass-, mid- and high-frequency spectrum

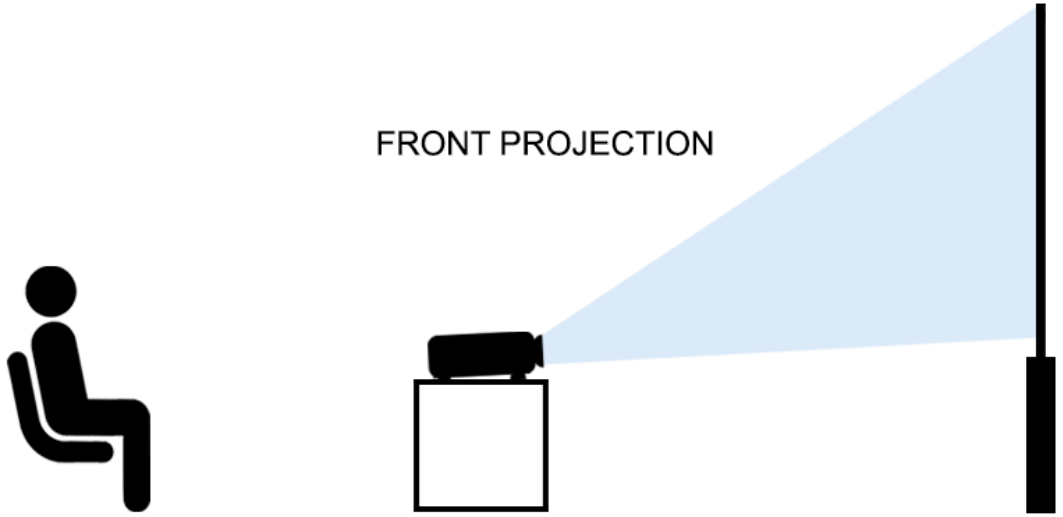
Used for reproduction of the mid- and high-frequency spectrum



1. A compression driver (A) in a horn loudspeaker consists of a metal diaphragm (blue) vibrated by the audio signal current in a coil of wire (red) between the poles of a cylindrical magnet (green). The sound waves pass out through an acoustic horn (B).

Front or rear projected display devices:

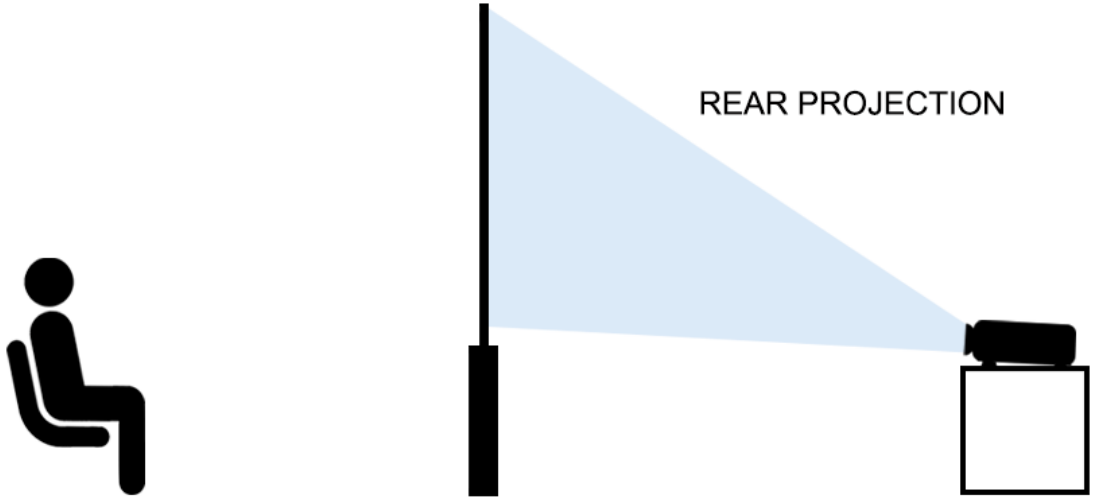
The viewing image for the audience is projected onto a front screen which reflects the image. *Front projection* is less expensive and requires much less space.



Source: https://blog.byjus.com/the-learning-tree/science-feed/make-projector-at-home-diy/

Rear projected Display devices:

- Essentially light from a rear screen projector is transmitted through the screen, made of material that transmits the light with relatively little distortion.
- *Rear screen* projection is preferred in bright, well-lighted environments.



Source: https://blog.byjus.com/the-learning-tree/science-feed/make-projector-at-home-diy/

3. SPEAKERS AND DISPLAYS

Rear and front screen projections can use light-emitting diodes (LED), liquid crystal displays LCD, or digital light projector (DLP) technology.

Major types of video displays include the following:

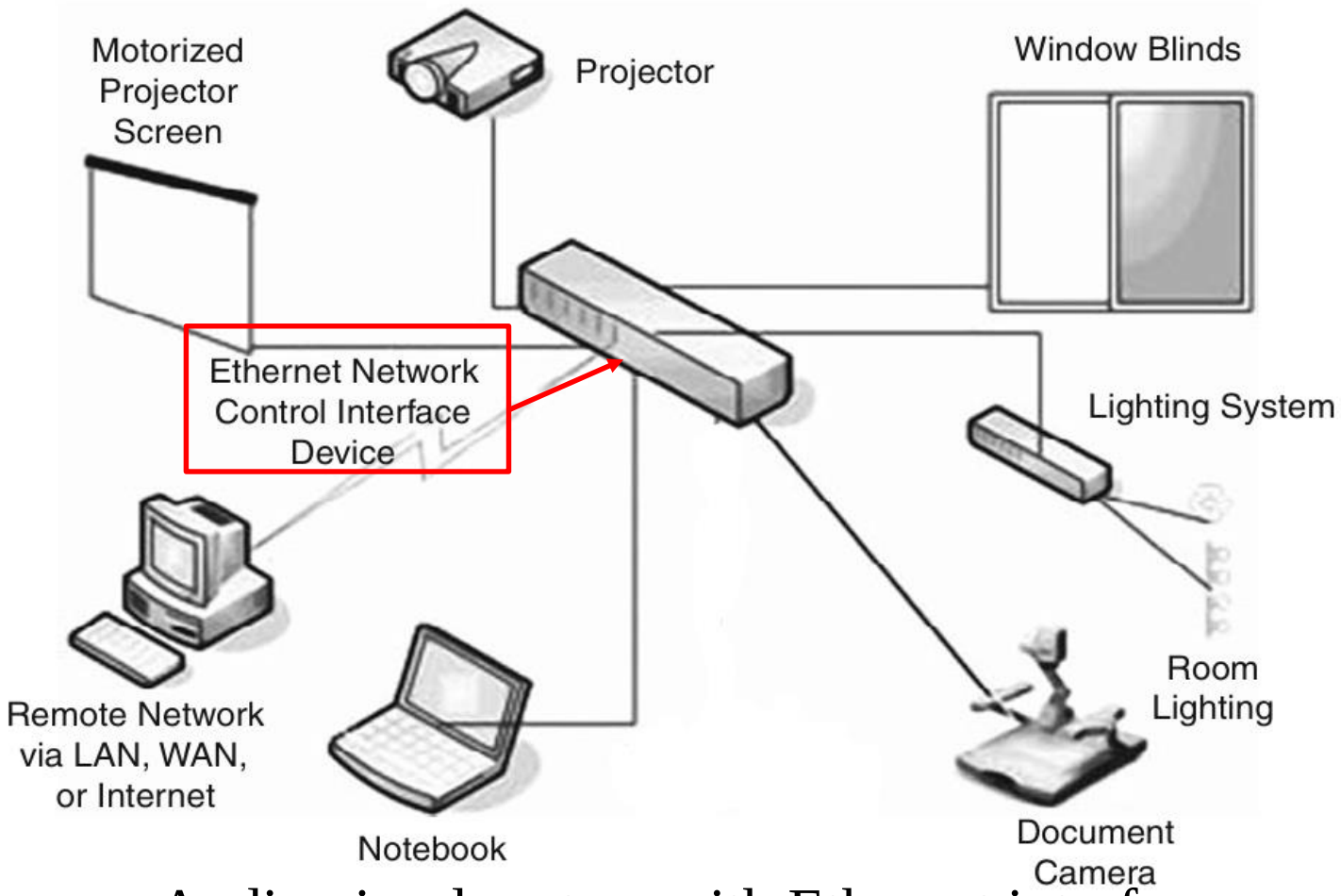
- Plasma Screen
- Digital Light Processing
- Liquid Crystal Displays
- Light-emitting diodes

4. AUDIO VISUAL CONTROL SYSTEMS

- Control systems provide user interfaces for managing and operating AV equipment efficiently, employing touch panels, remote controls, mobile apps, or centralized software platforms for adjusting settings, switching sources, and controlling multimedia content.

- Components may connect to an “Ethernet interface” which interfaces the various inputs and outputs from multiple components onto a standard Ethernet IP network.

4. AUDIO VISUAL CONTROL SYSTEMS

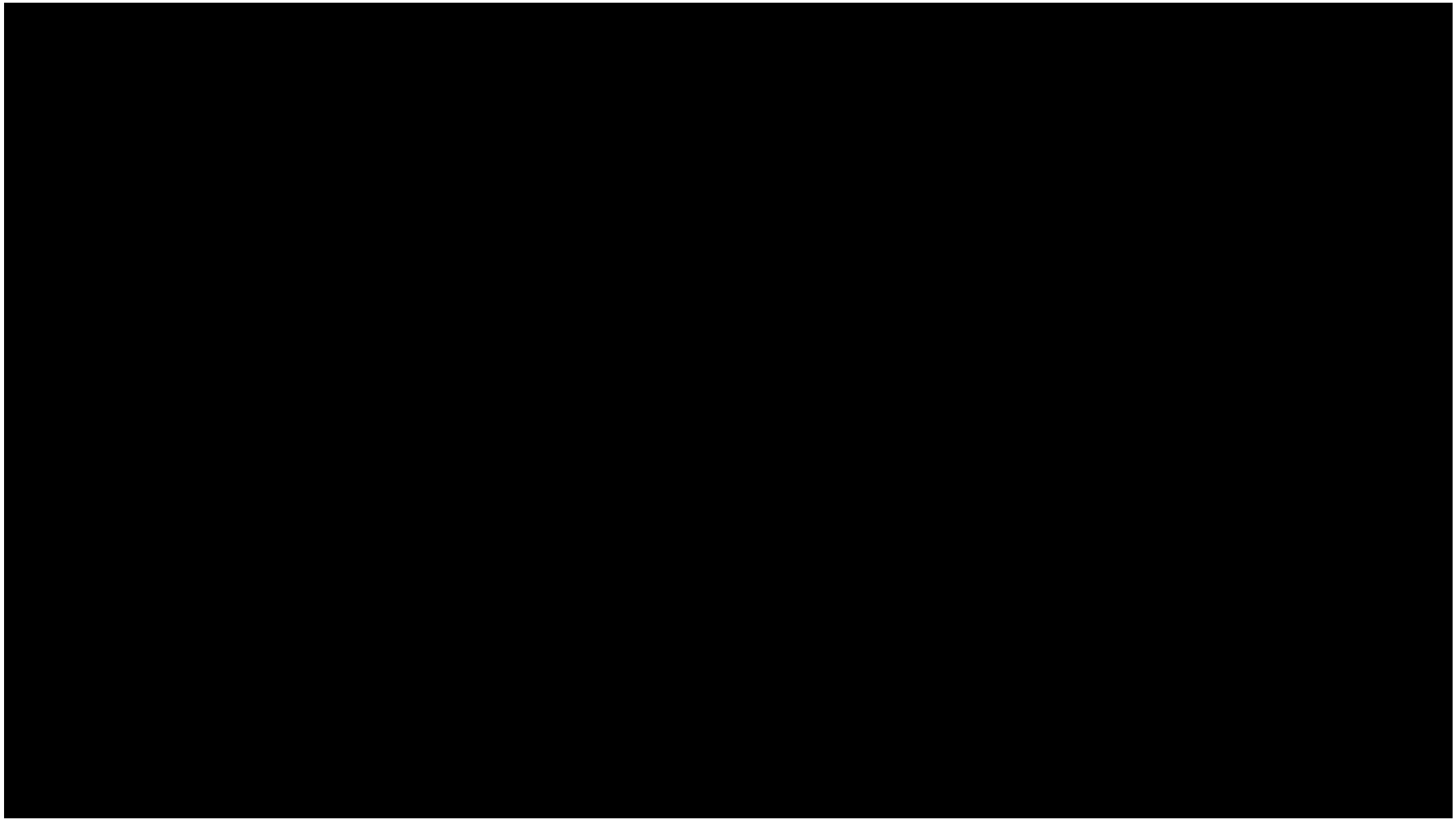


Audio visual system with Ethernet interface

Refer: Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

SUMMARY

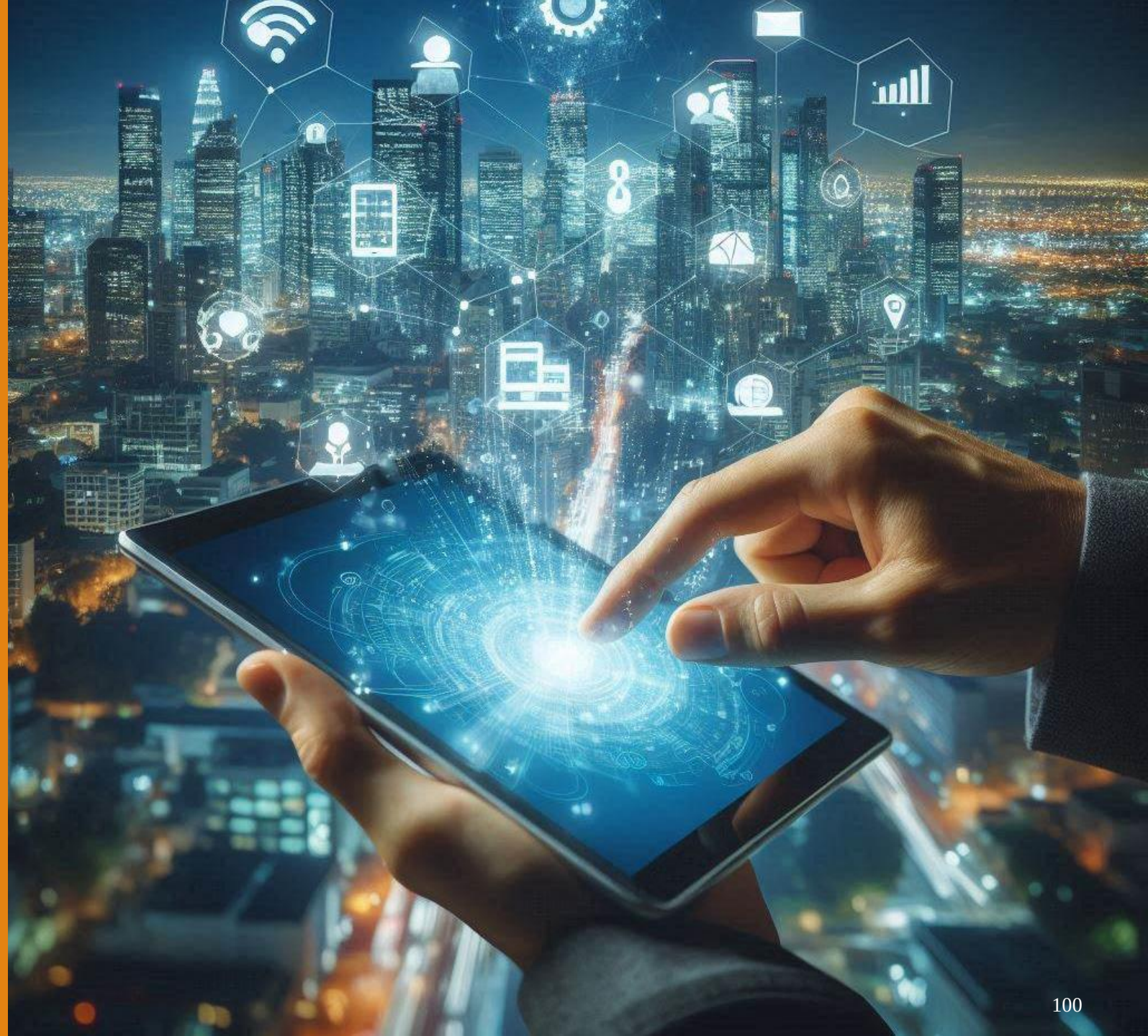
- Audio visual systems are essential tools that enhance the quality and effectiveness of communication, education, and entertainment.
- It enhances experiences by combining superior visual and audio elements, making it essential in various sectors from entertainment to business.
- As technology continues to advance, AV systems will become even more integral to our daily lives, providing innovative solutions and improving our interactions with the digital world.



https://www.youtube.com/watch?v=zcjiTFgKQls

LECTURE - 14

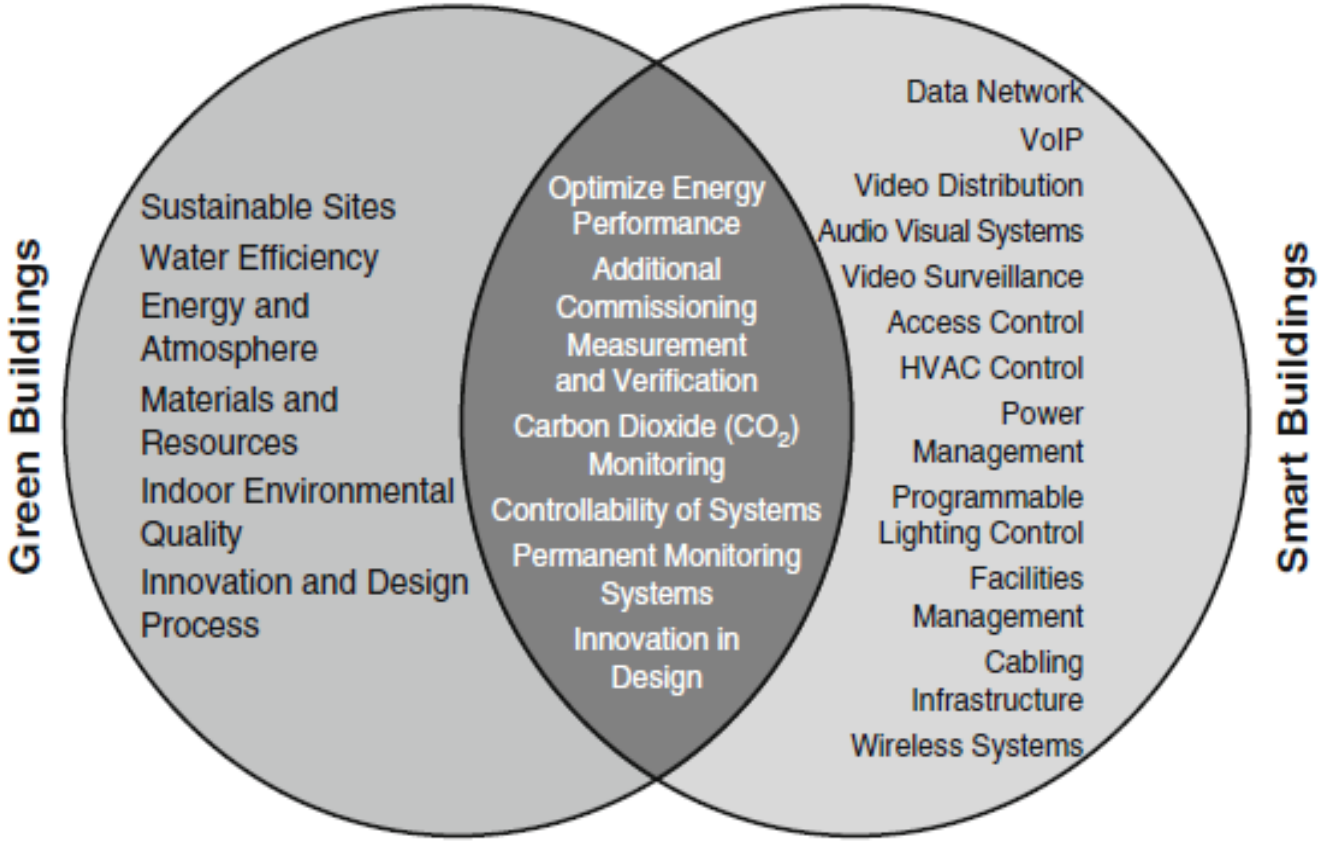
Energy and sustainability



Overview

- Integrating a building's technology systems and constructing a sustainable or "green" building have much in common.
- Green buildings are about resource efficiency, life-cycle effects and building performance.
- Smart buildings, whose core is integrated building technology systems, are about construction and operational efficiencies and enhanced management and occupant functions.

SMART AND GREEN BUILDINGS - COMMONALITY



Refer: Smart building systems for Architects, owners and builders, Elsevier, Butterworth-Heinemann Publications, 2010

CODES FOR GREEN BUILDINGS

- In India, the Green Building Code is a medley of codes and standards contained in the State by-laws, the National Building Code (NBC), and the Energy Conservation Building Code (ECBC).
- It includes norms set by ratings programs such as Leadership in Energy and Environmental Design-India (LEED-India), standards and guidelines for the Residential Sector by the Indian Green Building Council (IGBC), TERI-GRIHA, and other such certifications.

CODES FOR GREEN BUILDINGS

National Building Code (NBC):

- The Bureau of Indian Standards has developed the National Building Code (NBC) in the 1980s that guides municipalities and development authorities on building by-laws.
- In the latest version, the code provides guidance on aspects of energy conservation such as daylight integration, artificial lighting requirements, and HVAC design standards.

CODES FOR GREEN BUILDINGS

- Basic and general guidelines for efficient energy usage in the NBC do exist but they are merely guidelines.
- In the latest version of NBC i.e. 2016 version - Guidance has been provided for making buildings and built environment energy efficient and environmentally compatible, through the newly introduced and updated chapter on sustainability, Part 11 ‘Approach to Sustainability’.

Source: *https://www.bis.gov.in/index.php/standards/technical-department/national-building-code/*

CODES FOR GREEN BUILDINGS

Energy Conservation Building Code (ECBC):

- The Energy Conservation Building Code (ECBC) was launched in May 2007 by the Bureau of Energy Efficiency (BEE), Ministry of Power, Government of India.
- ECBC sets minimum energy standards for commercial buildings, with the objective of enabling energy savings of between 25 and 50% in compliant buildings.

CODES FOR GREEN BUILDINGS

- The code is applicable to commercial buildings like hospitals, hotels, schools, shopping complexes, and multiplexes which have a connected load of 100 kW or more, or contract demand of 120 kVA or more.
- The ECBC in India focuses on six key components of building design, including the envelope (walls, roofs, windows), lighting systems, HVAC (Heating, ventilation, and air conditioning) systems, and electrical power systems.

Refer Document: *https://beeindia.gov.in/sites/default/files/BEE_ECBC%202017.pdf*

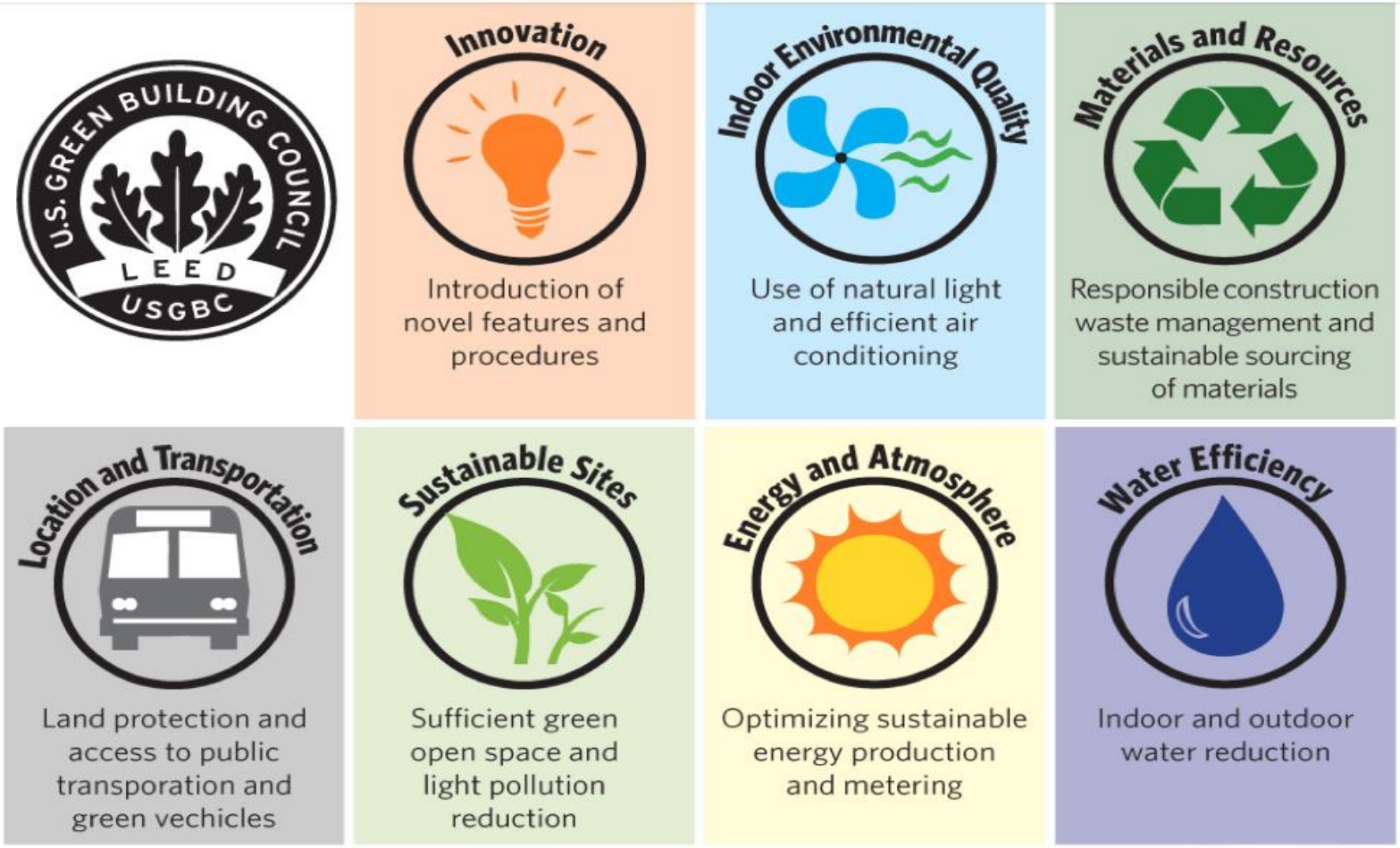
GREEN BUILDING CERTIFICATION

- LEED certification is a rating system created by the U.S. Green Building Council (USGBC) to evaluate the energy and sustainability aspects of a building.
- LEED is credit-based, giving points for certain eco-friendly measures that are taken during the construction and use of a building.

GREEN BUILDING CERTIFICATION

- Various LEED rating systems have been developed for different building types, sectors and project scopes.
- Included are LEED for Core and Shell, New Construction, Schools, Neighborhood Development, Retail, Healthcare, Homes, and Commercial Interiors.

LEED CERTIFICATION REQUIREMENT



Source: www.usgbc.org

LEED CERTIFICATION REQUIREMENT

Four Levels of LEED Certification

1. The Certified level is the entry-level certification and indicates basic adherence to LEED standards.

A project must earn a minimum number of points across various sustainability categories, including energy efficiency, water conservation, materials selection, indoor environmental quality, and innovation.

CERTIFIED
40 - 49 POINTS

LEED CERTIFICATION REQUIREMENT

2. The Silver level represents a higher level of sustainability performance than Certified.

Silver-certified projects typically demonstrate a more comprehensive implementation of green building strategies and technologies, resulting in greater environmental benefits.

SILVER

50 - 59 POINTS

LEED CERTIFICATION REQUIREMENT

3. The Gold level signifies exceptional sustainability performance and leadership in green building practices.

Silver-certified projects typically demonstrate a more comprehensive implementation of green building strategies and technologies, resulting in greater environmental benefits.

GOLD

60 - 79 POINTS

LEED CERTIFICATION REQUIREMENT

4. The Platinum level represents the highest level of LEED certification and is awarded to projects that demonstrate unparalleled sustainability excellence and innovation.

Platinum-certified projects showcase cutting-edge green building strategies, state-of-the-art technologies, and a deep commitment to environmental stewardship.

PLATINIUM
80+ POINTS

Smart Buildings and Green Buildings

- Integrated building technology systems align and support a green building approach.
- A smart building approach facilitates meeting or exceeding the technical requirements of the credits and points of the LEED rating system.

LEED CERTIFICATION RATING SYSTEM

Technical requirements of the LEED rating	Description	Points
Water efficient landscaping	To limit or eliminate the use of water for landscape irrigation by using electromechanical clocks to microprocessor-based controllers.	2-4
Fundamental commissioning of building energy systems & enhanced commissioning	• Quality assurance based process/ • Commissioning activities shall completed for HVAC, Lighting controls, Alternative energy systems, Water systems, and Power management systems	2

LEED CERTIFICATION RATING SYSTEM

Technical requirements of the LEED rating	Description	Points
Minimum energy performance & optimum energy performance	• An IP network is able to provide power-over-Ethernet (POE) to a range of “plug-load” devices. • Process loads must be identical for both the baseline building performance rating and the proposed building performance rating (“exceptional calculation method” ANSI/ASHRAE/IESNA Standard 90.1-2007 G2.5, to document measures that reduce process loads.)	1-19

LEED CERTIFICATION RATING SYSTEM

Technical requirements of the LEED rating	Description	Points
Measurement and verification	IPMVP is a guidance document that addresses determining and documenting savings resulting from energy-efficiency projects.	3
Outdoor air-delivery monitoring	• Monitoring system that can provide data on the ventilation of spaces that can be used to adjust the HVAC system • System will monitor air flow and carbon dioxide (CO2) levels	1

LEED CERTIFICATION RATING SYSTEM

Technical requirements of the LEED rating	Description	Points
Controllability of systems-lighting and thermal comfort	The capability of controlling the lighting, temperature, and ventilation of their space when building affords individual occupants or specific groups.	1
Innovation in design	There are two ways credits can be achieved • A new energy or sustainability strategy is proposed that is not addressed in current LEED strategies. • When the project has exceeded the credit requirements by 100% or has met the next incremental percentage threshold of an existing credit	1-5

LEED Green Ratings:

- In India, LEED certification is a widely recognized global certification for green building projects.
- Post June 5, 2014, Green Business Certification Inc. (GBCI) is India's only body responsible for providing LEED certification.

GREEN BUILDING CERTIFICATION IN INDIA

- Standards have been customized according to Indian conditions, in terms of the design, construction and operation of buildings for environment-friendly performance
- There are currently three LEED certification types under the IGBC: LEED India for New Construction (NC), Core and Shell (CS), Existing Buildings (EB).

Source: https://media.biltrax.com/what-are-the-green-building-rating-systems-in-india/

GRIHA (The Green Rating for Integrated Habitat Assessment):

- Endorsed by the Ministry of New and Renewable Energy, Government of India as of November 1 2007, GRIHA is a five star rating system for green buildings which emphasizes on passive solar techniques for optimizing indoor visual and thermal comfort.
- The rating applies to new building stock – commercial, institutional, and residential of varied functions.

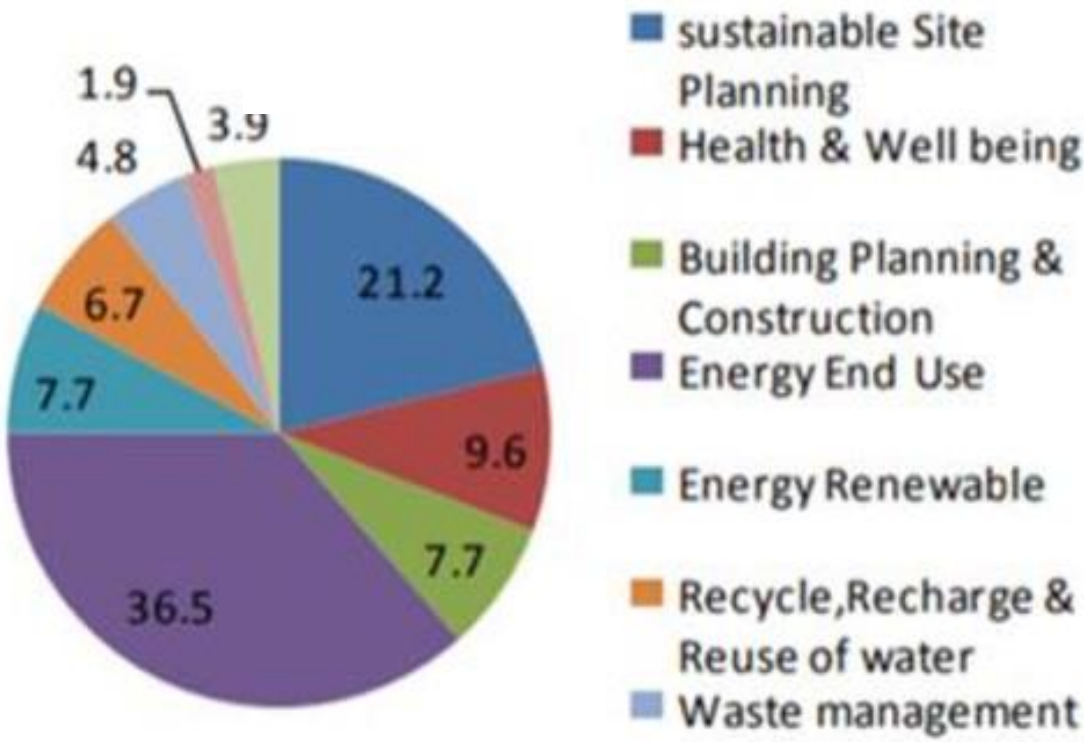
Source: Manual VolI PDF (WWW.GRIHAINDIA.ORG)

GREEN BUILDING CERTIFICATION IN INDIA

GRIHA Rating system based on 8 major categories

1) Sustainable Site Planning
2) Health & Well-being
3) Building Planning & Construction
4) Energy End Use
5) Energy Renewable
6) Recycle, Recharge & Reuse of water
7) Waste management
8) Indoor air quality.

GREEN BUILDING CERTIFICATION IN INDIA



GRIHA points category

*Source: **IRJET-V6I2370** PDF (WWW.IRJET.NET)*

POINTS	GRIHA RATING
50-60	★
61-70	★★
71-80	★★★
81-90	★★★★
91-100	★★★★★

GRIHA evaluation system

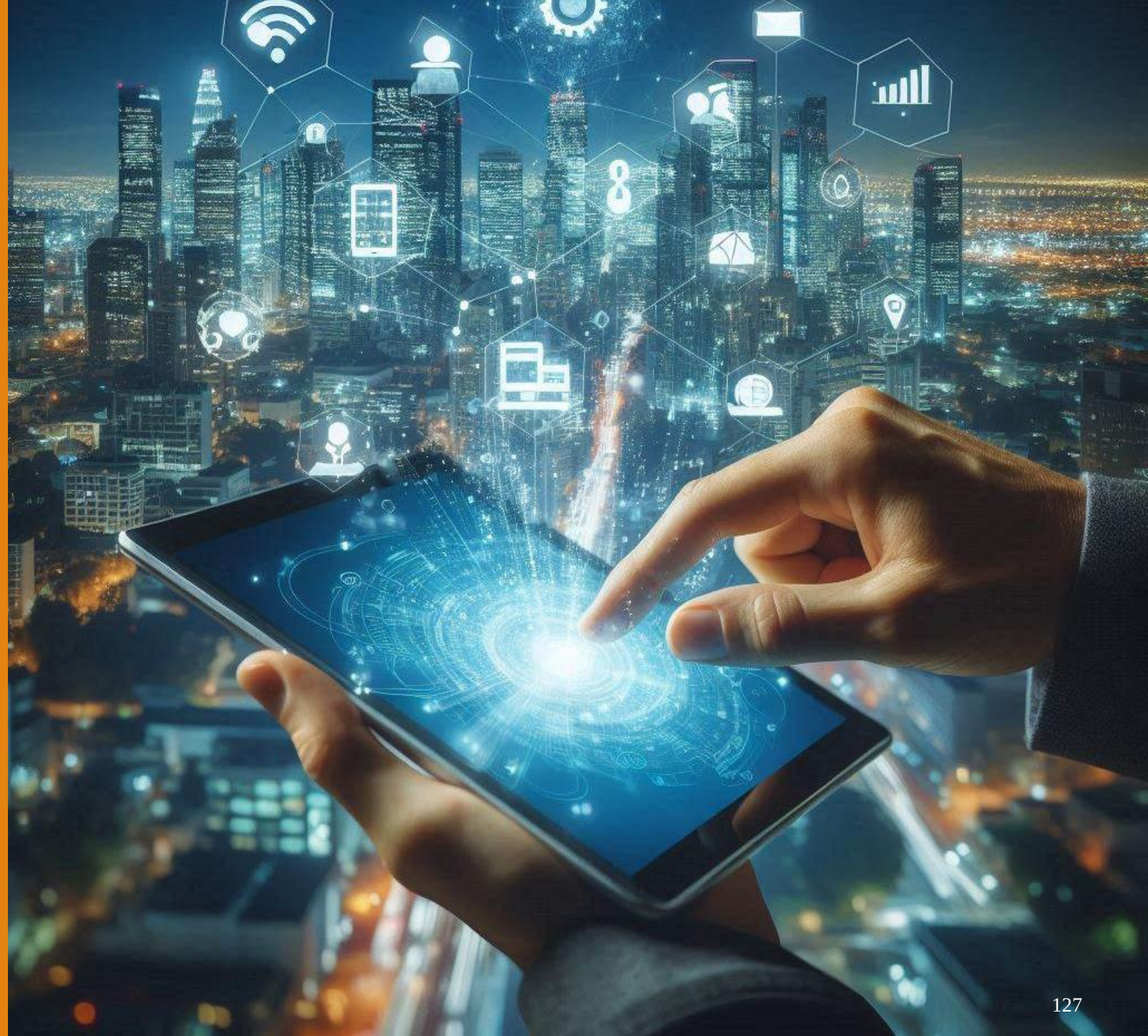
Source: Manual VolI PDF (WWW.GRIHAINDIA.ORG)

SUMMARY

- By adopting green building practices, utilizing innovative technologies, and embracing a holistic approach to building design and operation, we can create structures that not only reduce environmental impact but also provide economic benefits and improve occupant well-being.
- The path to sustainable buildings involves overcoming initial costs and technical complexities, but the long-term advantages make it a worthwhile and necessary endeavor.

https://youtu.be/DRO_rIkywxQ

Case Studies



CASE STUDY

Overview

- In today's buildings, it is not sufficient to merely integrate the air-conditioning with lighting, or physical security with access control.
- Going beyond randomly chosen systems, advanced technological capabilities are now enabling components such as HVAC, lighting, security cameras, fire alarms, electrical, building-access control systems as well as voice and data communication to be integrated and controlled intelligently and seamlessly from a single network.

CASE STUDY 1

Ave Maria University

- The campus is situated on about 1,000 acres, with the initial building phase consisting of 500,000 square feet of facilities, serving nearly 500 students and 200 faculty and staff.
- The campus development has been conceived with state-of-the-art design elements and integration of technological systems to result in operational efficiencies and cost savings.

CASE STUDY 1

- To the owners, converging various technologies made sense from both a construction and ongoing operations standpoint and this proved to be a key deciding factor in choosing single party technology contracting to design, install and oversee the management of multiple systems throughout the campus.

- From IT to facility operations, the infrastructure encompasses nearly 23 systems that were proposed to be converged and made interoperable on a single IP network.

Refer: Bright Green Buildings: The Convergence of Green and Intelligent Buildings

CASE STUDY 1

Project objectives:

- Achieve cost reductions by avoiding the installation of separate and proprietary mechanical, electrical and communications systems.
- Minimize human resource dependency by reducing physical monitoring and maintenance of separate systems across the 908-acre campus.
- Monitor utility usage and expenses, avoid power spikes and excess supply issues while seeking financial reconciliation for discrepancies with utilities.
- Manage nearly 23 facility and IT systems on campus from a single network operations center.

CASE STUDY 1

Key challenges:

- Capital Cost Outlay: Managing budgets for mechanical and electrical installations for each separate system, both for facility operations and IT, and carefully allocating costs for each system's network of wires and cabling as well as proprietary protocols for control and communications.
- Anticipation of Incremental Manpower Costs: Human resource allocation across departments such as IT, Life Safety and Operations to monitor and manage tasks for each isolated system on campus meant an incremental cost exposure for hiring technically qualified manpower.

CASE STUDY 1

- Management Challenges: With separate external communications connections for each system under a conventional setup, the operations team would be required to manually monitor the operations of each system. This challenged the ability to detect problems in real time and avoid damages and secondary impacts.

CASE STUDY 1

Solutions:

- In view of the above challenges, the project team at Ave Maria University realized that the best way to achieve the end objectives was to centralize operations and management of all systems over a single network.
- Towards this objective the project team decided to work with *Smart Buildings* – a professional engineering and consulting services company, engaged in the area of designing integrated building technology systems for cost effectiveness and functionality.

CASE STUDY 1

- Buildings designed a solution that encompassed lighting control, power management, HVAC control, data networks, voice networks, audio-visual systems, video distribution, video surveillance, access control and facilities management
- To put the design concept into action, Smart Buildings worked closely to assemble industry experts who would be delegated with specific functionalities, based on the expertise they bring to the table.

CASE STUDY 1

Partnering for efficient delivery:

- The partnership model for service delivery was based on the ‘Technology Contracting’ approach as opposed to traditional contracting with multiple subcontractors installing separate proprietary systems.

- This approach involved entrusting the responsibility for planning, designing, installing, integrating, commissioning and servicing technology systems throughout the project footprint to a single provider.

CASE STUDY 1

- Johnson Controls Inc, who was appointed *technology contractor* for the project, supervised the design and installation of an IP infrastructure and the technology that resides on the network.
- Cisco Systems Inc who was appointed information network partner for the project, built a 24-hour network operations center on campus that controls and monitors all university communications and building operations from one location.

CASE STUDY 1

- *The Metasys system*[1] helped operators manage and mechanize the campus's HVAC setup, chiller plant, indoor air quality, air flow, lighting and lavatories, in addition to carrying out power management and asset tracking.
- In lieu of wiring and cabling for each proprietary system, a single IP network running over Ethernet cabling was installed and all systems, devices, and sensors interface to this Ethernet network.

1. Building Automation System Software that Delivers Complete Control

CASE STUDY 1

Reaping a smart harvest:

- The state-of-the-art, nine-building campus has a fiber-optic network that links each building, with Wi-Fi access throughout the campus.
- Maintenance is fully mechanized with temperature control and pressure readings all taken care of by staff members through simple mouse clicks and keyboard adjustments.
- There is a 900-square-foot network operations center with wall-mount LCD TVs, for the staff member to monitor the campus' IP surveillance cameras and view real-time data from building systems.

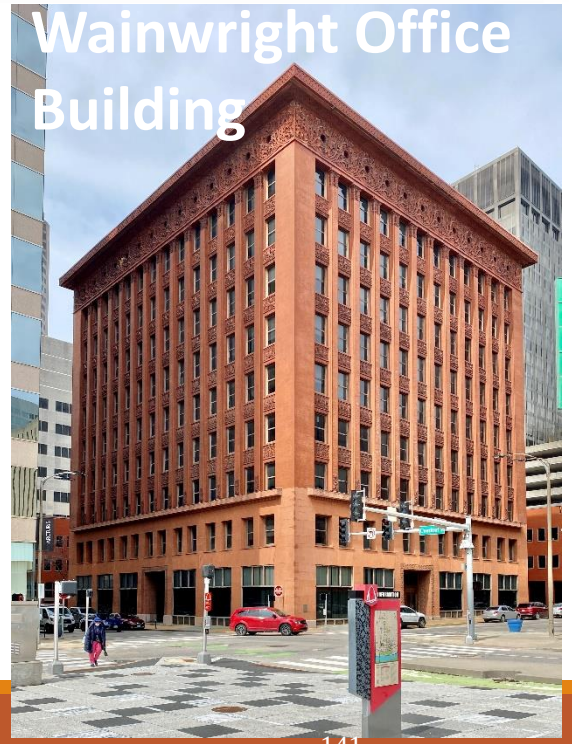
RESULTS

- Reduced staffing costs by enabling IT to assume tasks of building maintenance staff with an estimated US $350,000 savings annually in human resource costs.
- Saved over $1 million in building costs by eliminating the redundant wiring and cabling of multiple isolated building systems.
- Enabled significant efficiencies in utility usage with an estimated US $600,000 in annual savings.

CASE STUDY 2

State of Missouri

- The state owned or leased over $4 billion of real estate and was spending $300 million a year for acquisition, renovation and upkeep.
- The state also announced a plan to reduce building energy consumption by 15% by 2010.
- The goal was to address energy efficiency, cost of deferred maintenance, operating cost, space utilization, and asset management.



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CASE STUDY 2

The aim of this project was to bring rapid and quantifiable cost savings in five key areas, namely:

1. Utility bill management
2. Automated enterprise monitoring
3. Facilities communications infrastructure
4. Portal collaboration
5. Business process management implementation and improvement

CASE STUDY 2

- Johnson Controls played the role of the energy service company (ESCO) and accepted the performance risk to achieve the sustainable goals set forth by the State of Missouri.
- Its main focus was on the integration between building automation systems and a wireless communications backbone to help deliver a complete building information management system.

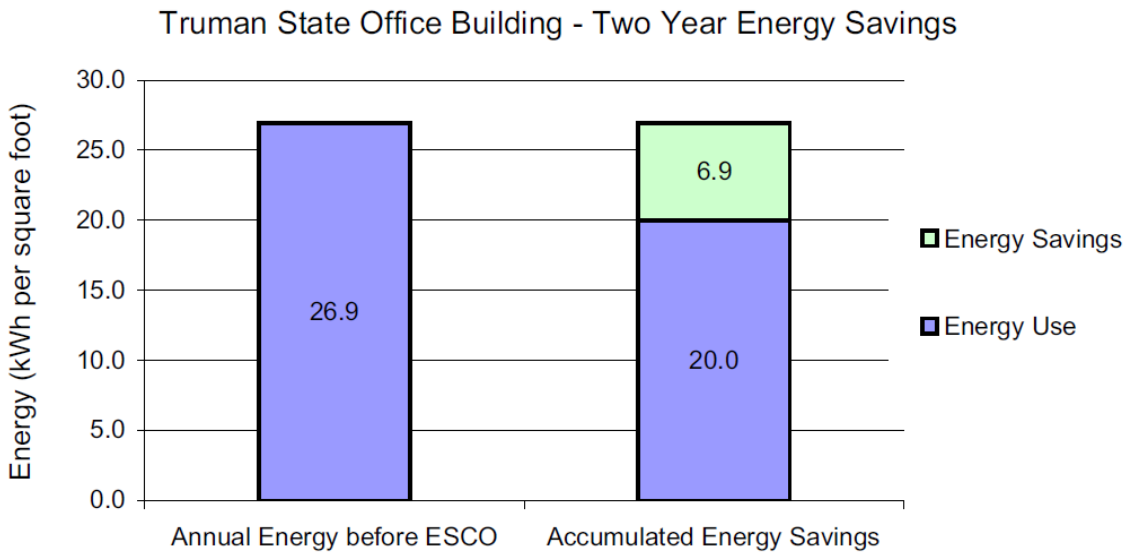
CASE STUDY 2

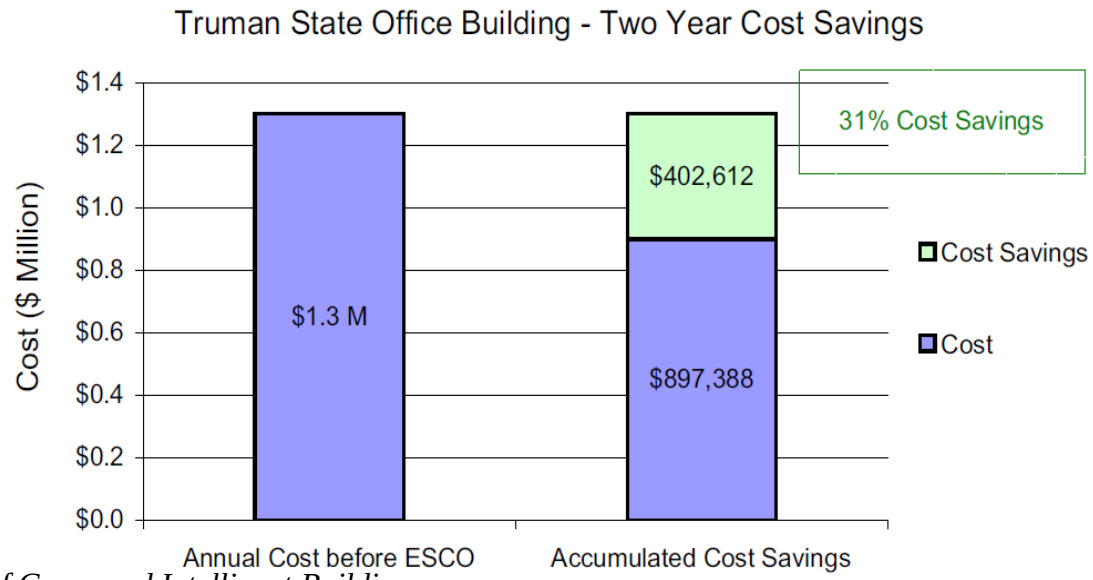
- The software solution provided by Gridlogix1 enabled the State of Missouri to integrate with existing control systems and applications to communicate together and help reduce energy, maintenance, compliance, and physical security cost.

1. GridLogix is a leading creator of XML Web Services based integration solutions for remote control and automation systems

RESULTS- PILOT 1

- Truman State Office Building, Jefferson City, Missouri
- 775,000 square feet - the State's biggest office building
- 31 percent energy savings after two years - almost double the guarantee

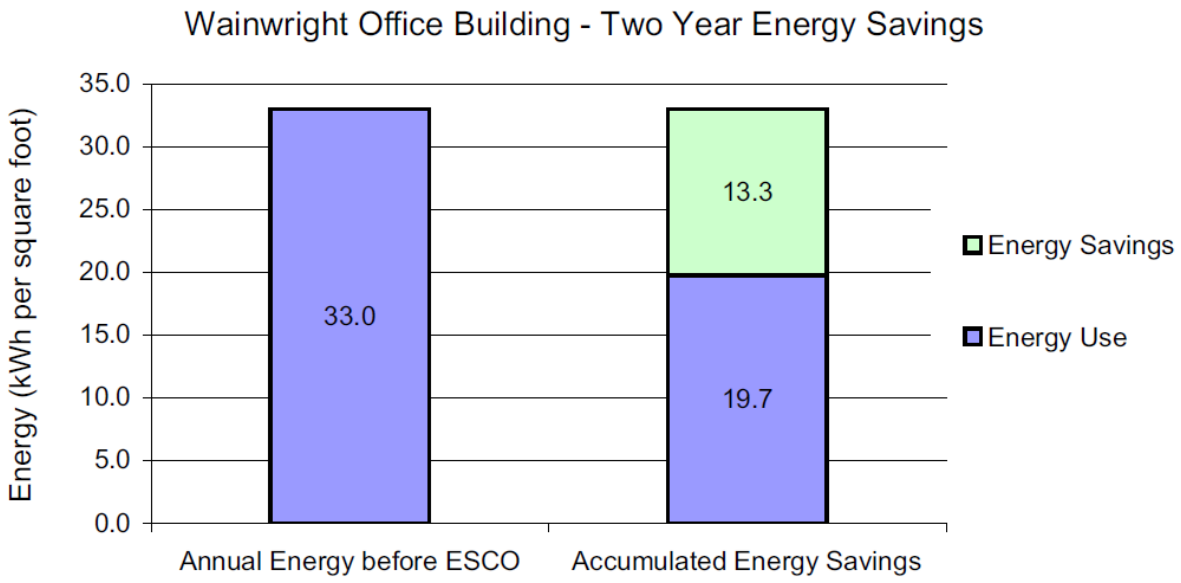


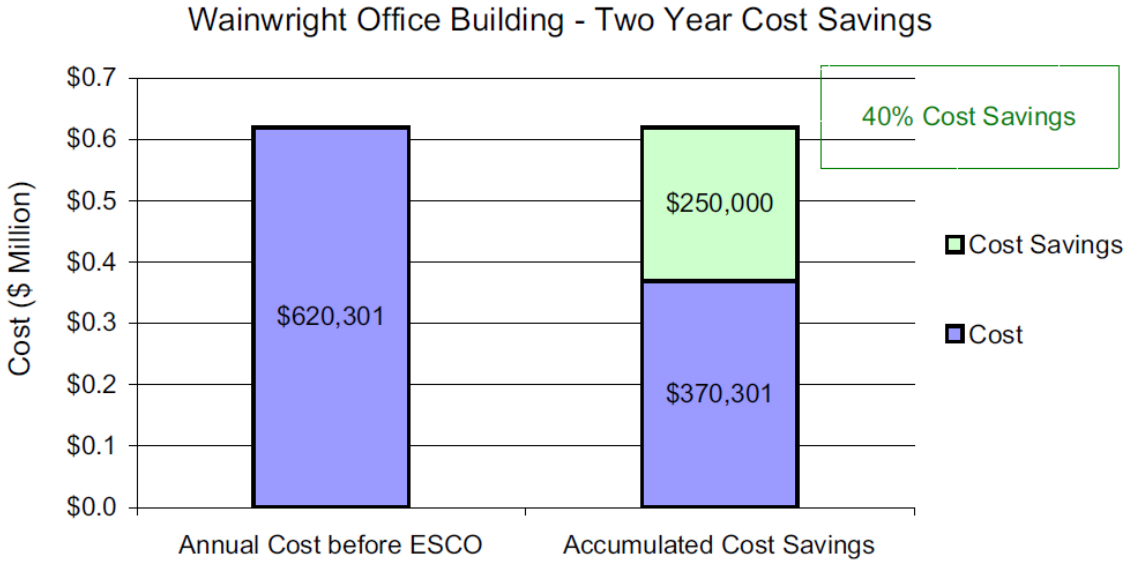


Source :Bright Green Buildings: The Convergence of Green and Intelligent Buildings

RESULTS- PILOT 2

- Wainwright Office Building, Downtown St. Louis, Missouri
- 234,000 square feet
- 40 percent energy savings after two years





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SUMMARY

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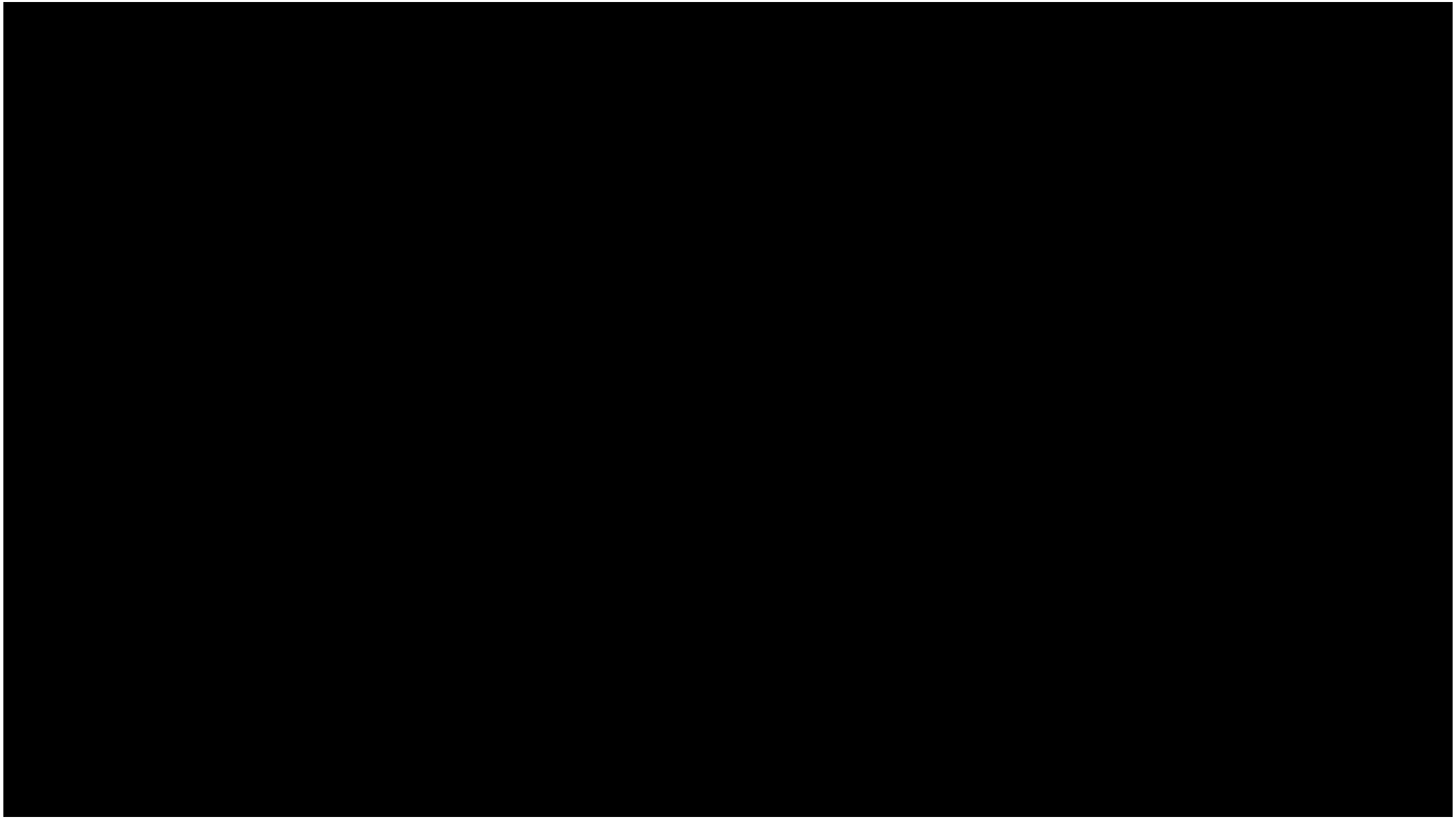

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